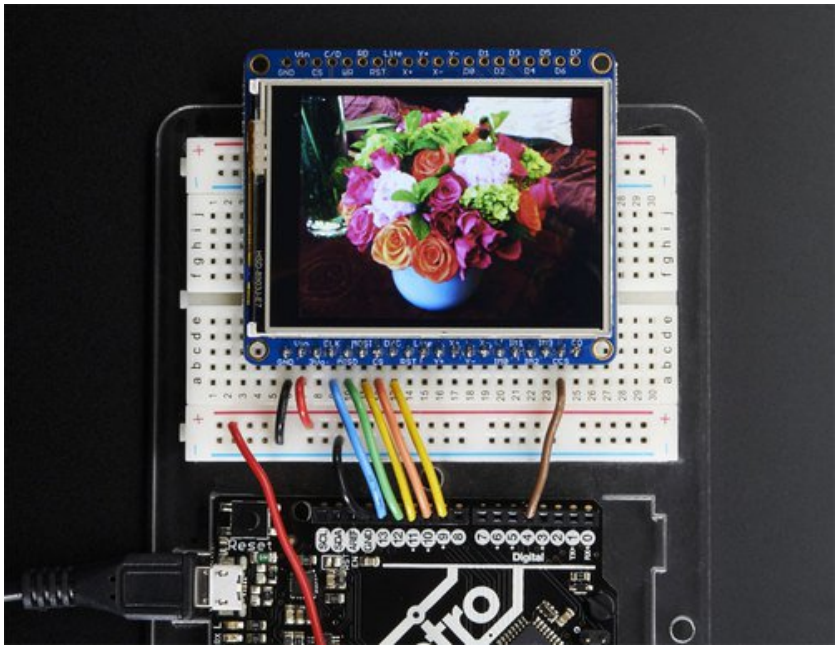
adafruit learning system

Adafruit 2.4" Color TFT Touchscreen Breakout

Created by lady ada



Last updated on 2017-06-26 03:55:28 AM UTC

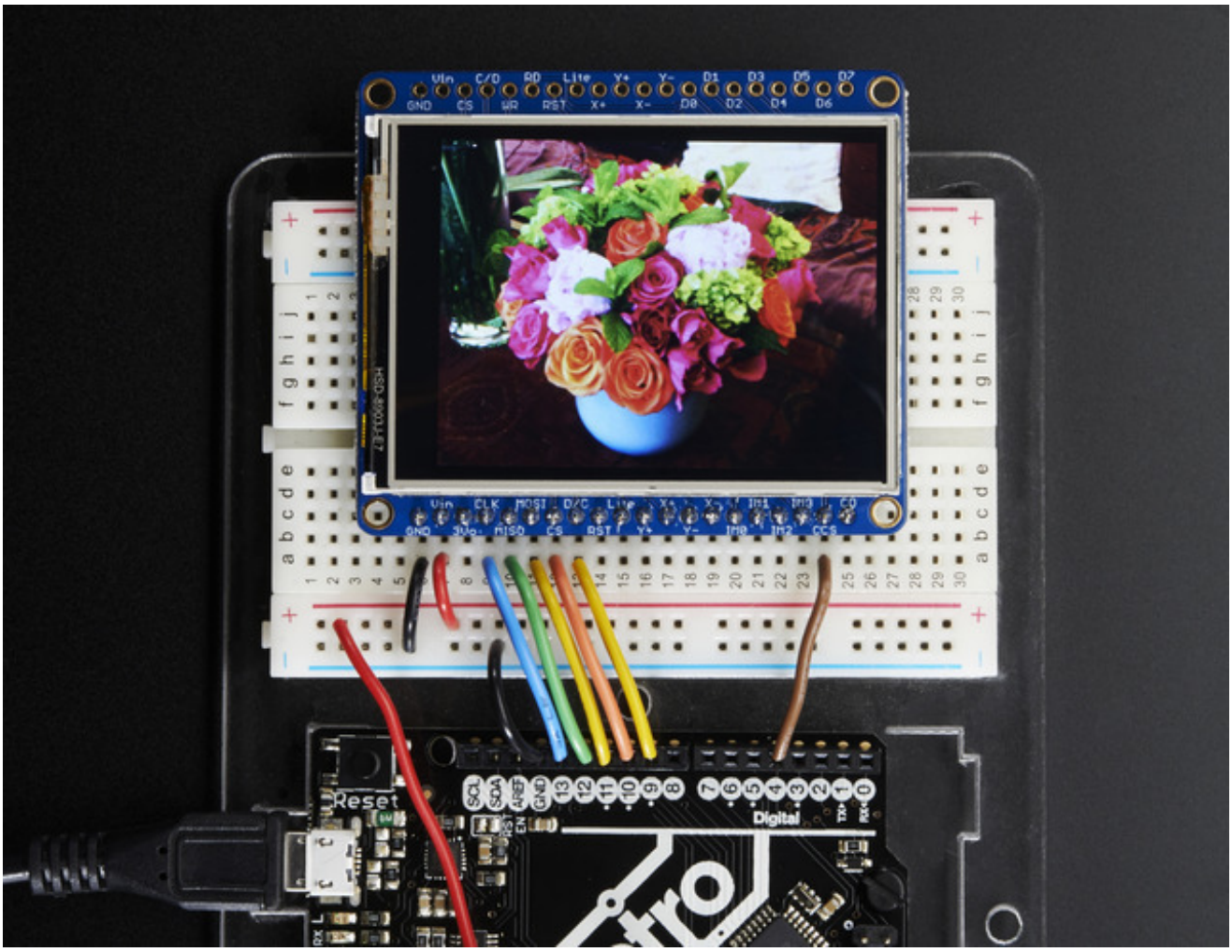
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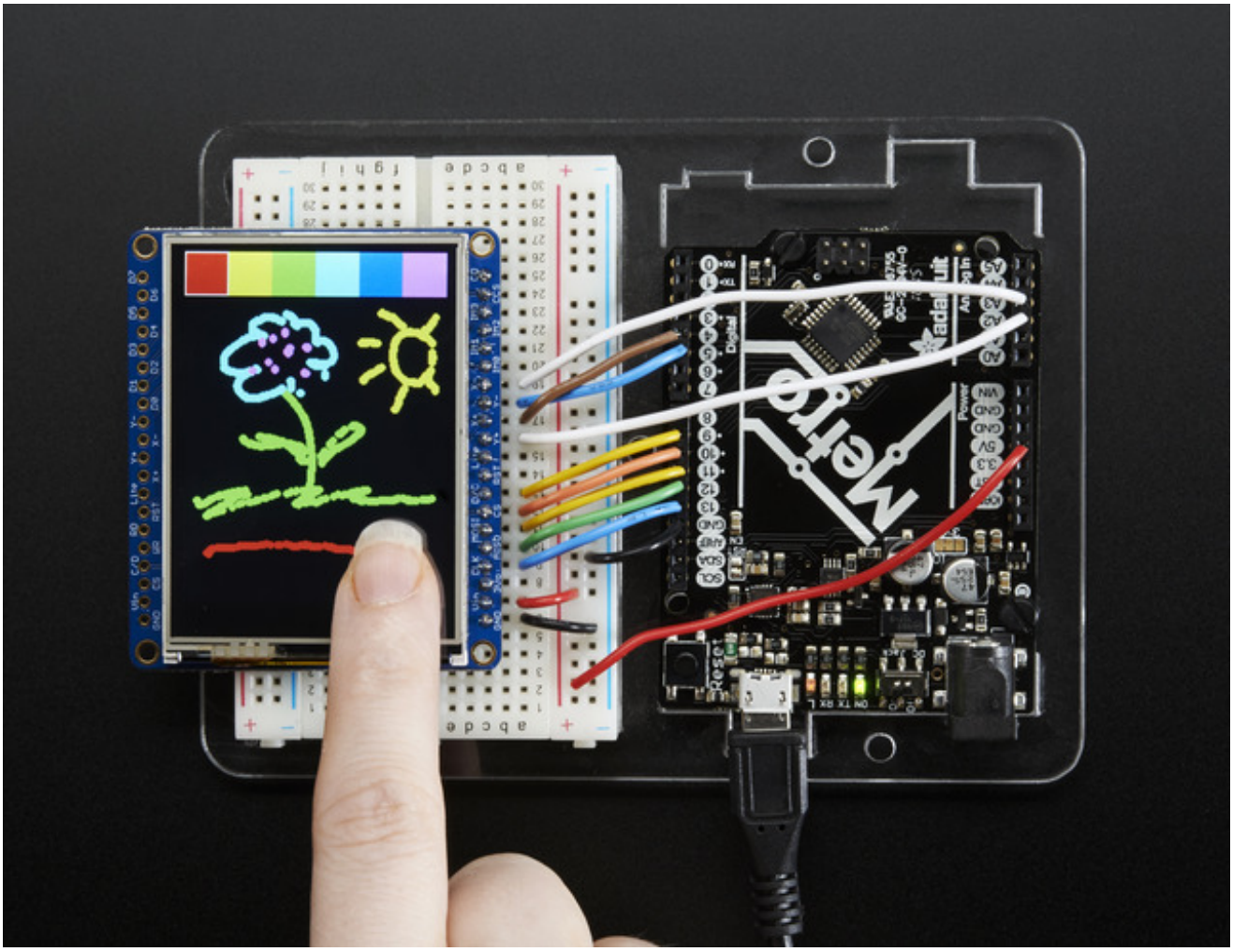
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Overview



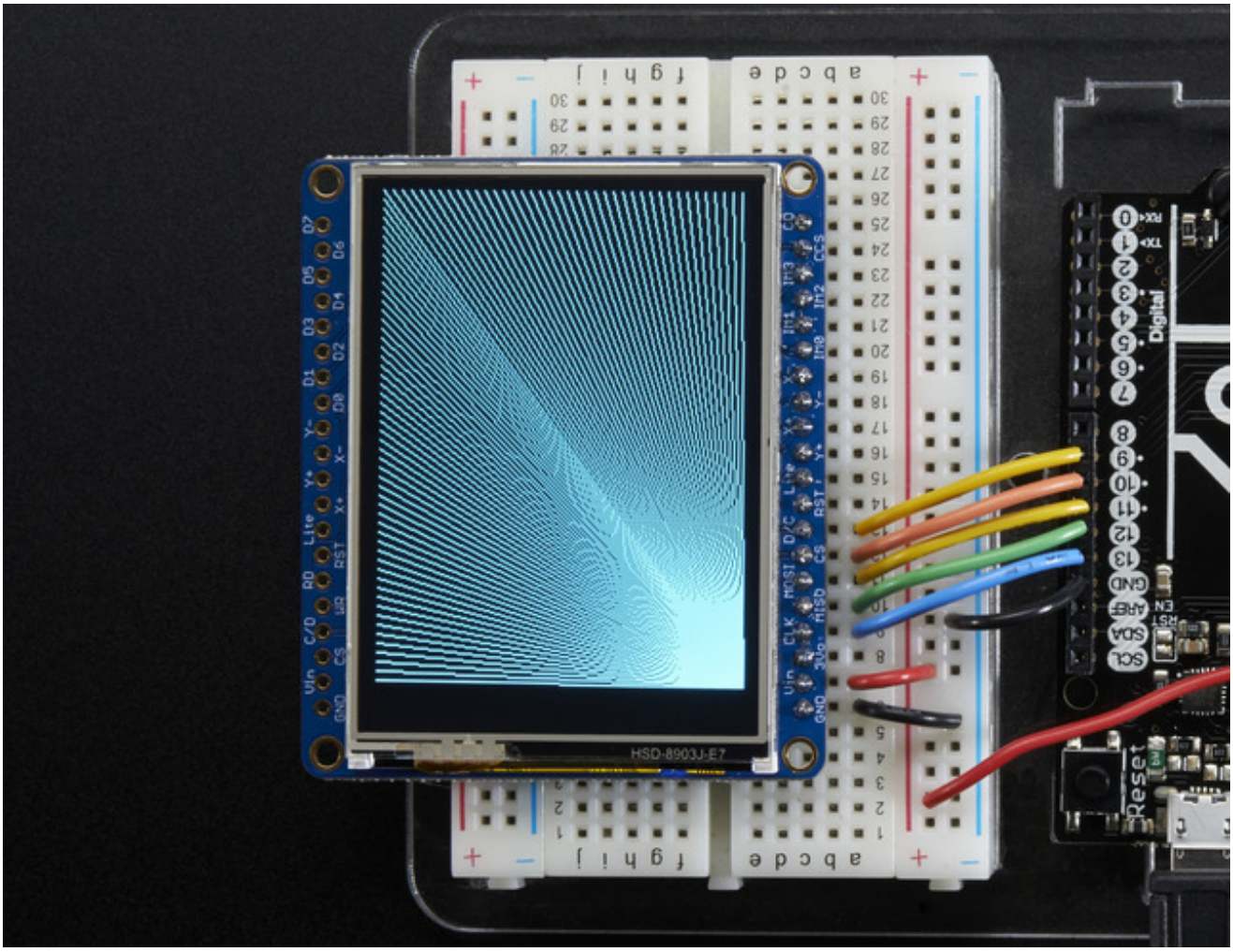
Add some jazz & pizazz to your project with a color touchscreen LCD. This TFT display is 2.4" diagonal with a bright (4 white-LED) backlight and it's colorful! 240x320 pixels with individual RGB pixel control, this has way more resolution than a black and white 128x64 display.

As a bonus, this display has a resistive touchscreen attached to it already, so you can detect finger presses anywhere on the screen.



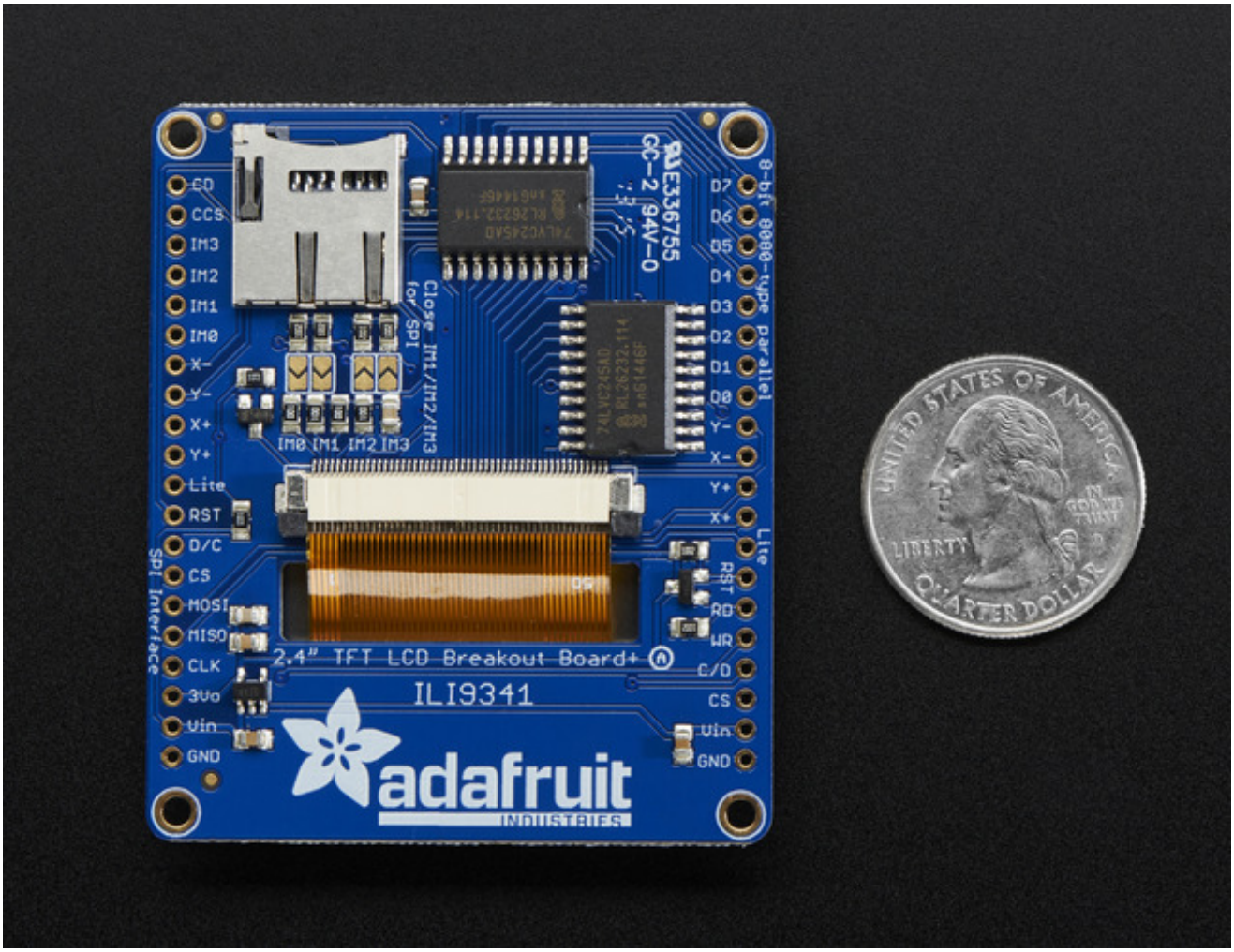
This display has a controller built into it with RAM buffering, so that almost no work is done by the microcontroller. **The display can be used in two modes: 8-bit or SPI.** For 8-bit mode, you'll need 8 digital data lines and 4 or 5 digital control lines to read and write to the display (12 lines total). SPI mode requires only 5 pins total (SPI data in, data out, clock, select, and d/c) but is slower than 8-bit mode.

In addition, 4 pins are required for the touch screen (2 digital, 2 analog) or [you can purchase and use our resistive touchscreen controller (not included) to use I2C or SPI](http://adafru.it/1571) (http://adafru.it/1571)



Of course, we wouldn't just leave you with a datasheet and a "good luck!". For 8-bit interface fans we've written a full open source graphics library that can draw pixels, lines, rectangles, circles, text, and more (http://adafru.it/aHk). For SPI users, we have a library as well (http://adafru.it/d4d), its separate from the 8-bit library since both versions are heavily optimized.

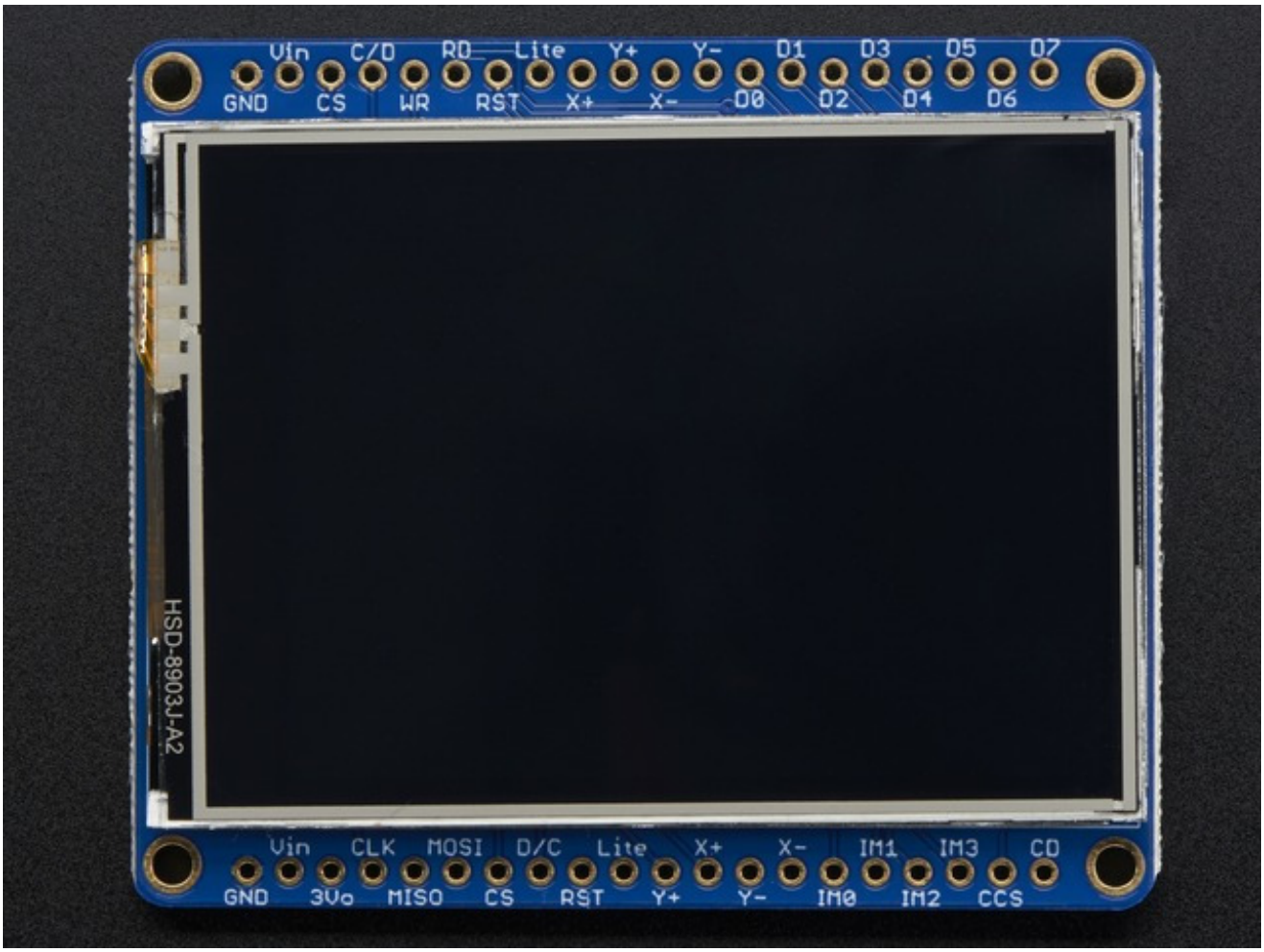
For resitive touch, we also have a touch screen library that detects x, y and z (pressure) (http://adafru.it/aT1) and example code to demonstrate all of it.


CD
CCS
IM3
IM2
IM1
IM0
X-
Y-
X+
Y+
Lite
RST
D/C
CS
MOSI
MISO
CLK
3Vo
Vin
GND
SPI Interface
IM0 IM1 IM2 IM3
Close IM1/IM2/IM3 for SPI
E336755
GC-2 94V-0
8-bit 8080-type parallel
D7
D6
D5
D4
D3
D2
D1
D0
Y-
X-
Y+
X+
Lite
RST
RD
WR
C/D
CS
Vin
GND
2.4" TFT LCD Breakout Board+
ILI9341
adafruit
INDUSTRIES
LIBERTY
QUARTER DOLLAR

Pinouts

The 2.8" TFT display on this breakout supports many different modes - so many that the display itself has 50 pins. However, we think most people really only use 2 different modes, either "SPI" mode or 8-bit mode (which includes both 6800 and 8080). Each 'side' of the display has all the pins required for that mode. You can switch between modes, by rewiring the display, but it cannot be used in two modes at the same time!

All logic pins, both 8-bit and SPI sides, are 3-5V logic level compatible, the 74LVX245 chips on the back perform fast level shifting so you can use either kind of logic levels. If there's data output, the levels are at at 3.3V



SPI Mode

This is what we think will be a popular mode when speed is not of the utmost importance. It doesn't use as many pins (only 4 to draw on the TFT if you skip the MISO pin), is fairly flexible, and easy to port to various microcontrollers. It also allows using a microSD card socket on the same SPI bus. However, its slower than parallel 8-bit mode because you have to send each bit at a time instead of 8-bits at a time. Tradeoffs!



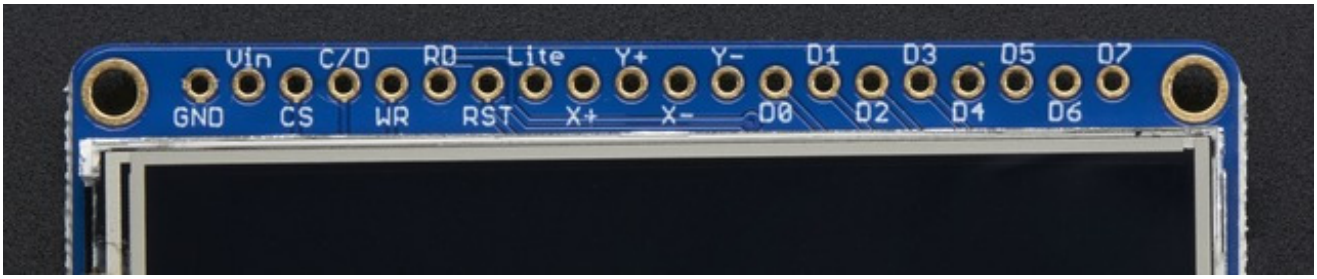
- **GND** - this is the power and signal ground pin
- **3-5V** / **Vin** - this is the power pin, connect to 3-5VDC - it has reverse polarity protection but try to wire it right!
- **3.3Vout** - this is the 3.3V output from the onboard regulator
- **CLK** - this is the SPI clock input pin
- **MISO** - this is the SPI Master In Slave Out pin, its used for the SD card mostly, and for debugging the TFT display. It isn't necessary for using the TFT display which is write-only
- **MOSI** - this is the SPI Master Out Slave In pin, it is used to send data from the microcontroller to the SD card and/or TFT
- **CS** - this is the TFT SPI chip select pin
- **D/C** - this is the TFT SPI data or command selector pin
- **RST** - this is the TFT reset pin. There's auto-reset circuitry on the breakout so this pin is not required but it can be helpful sometimes to reset the TFT if your setup is not always resetting cleanly. Connect to ground to reset the TFT
- **Lite** - this is the PWM input for the backlight control. It is by default pulled high (backlight on) you can PWM at any frequency or pull down to turn the backlight off
- **IM3 IM2 IM1 IM0** - these are interface control set pins. In general these breakouts aren't used, and instead the onboard jumpers are used to fix the interface to SPI or 8-bit. However, we break these out for advanced use and also for our test procedures
- **Card CS** / **CCS** - this is the SD card chip select, used if you want to read from the SD card.
- **Card Detect** / **CD** - this is the SD card detect pin, it floats when a card is inserted, and tied to ground when the card is not inserted. We don't use this in our code but you can use this as a switch to detect if an SD card is in place without trying to electrically query it. Don't forget to use a pullup on this pin if so!

Resistive touch pins

- **Y+ X+ Y- X-** these are the 4 resistive touch screen pads, which can be read with analog pins to determine touch points. They are completely separated from the TFT electrically (the overlay is glued on top) They can be used in 8-bit or SPI mode.

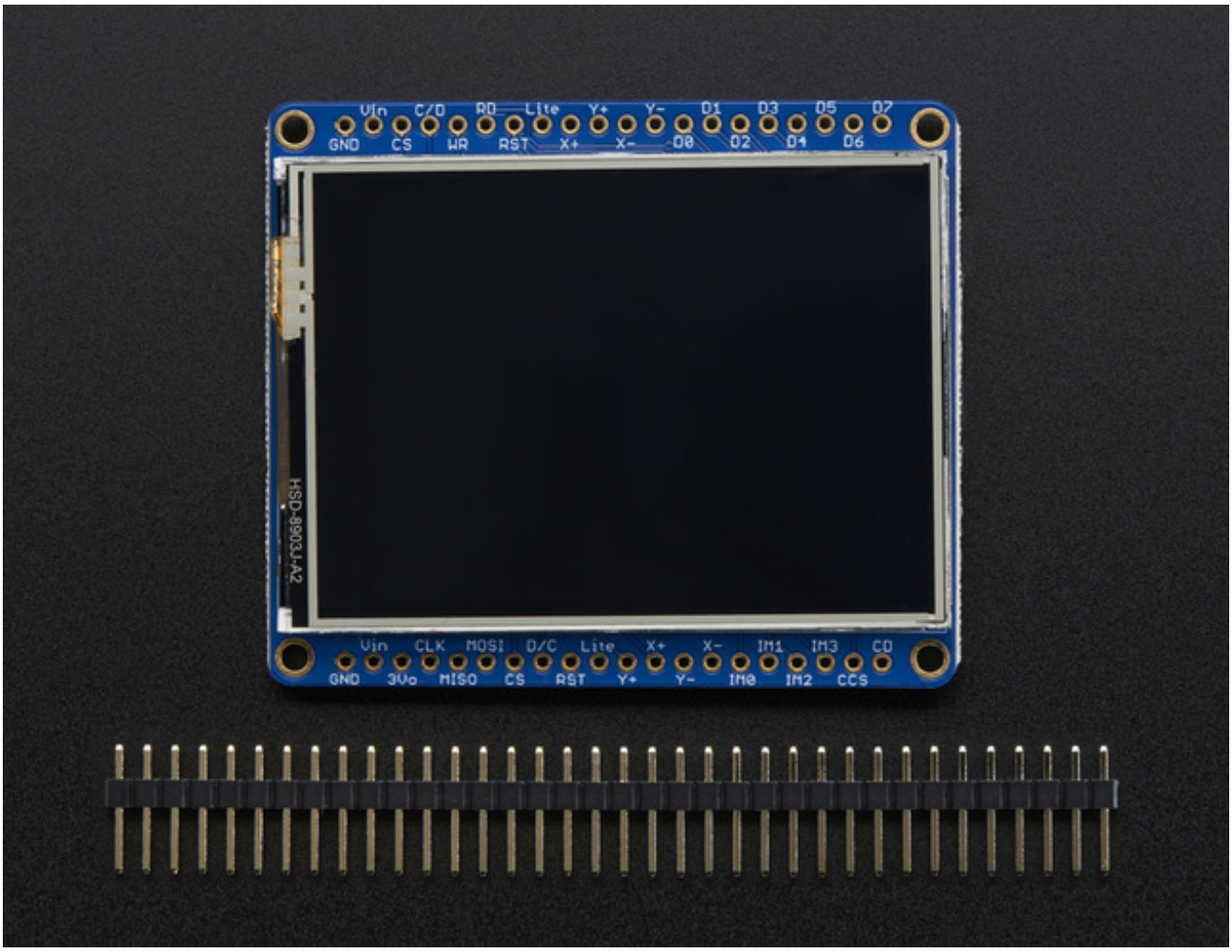
8-Bit Mode

This mode is for when you have lots of pins and want more speed. In this mode we send 8 bits at a time, so it needs way more pins, 12 or so (8 bits plus 4 control)! If your microcontroller

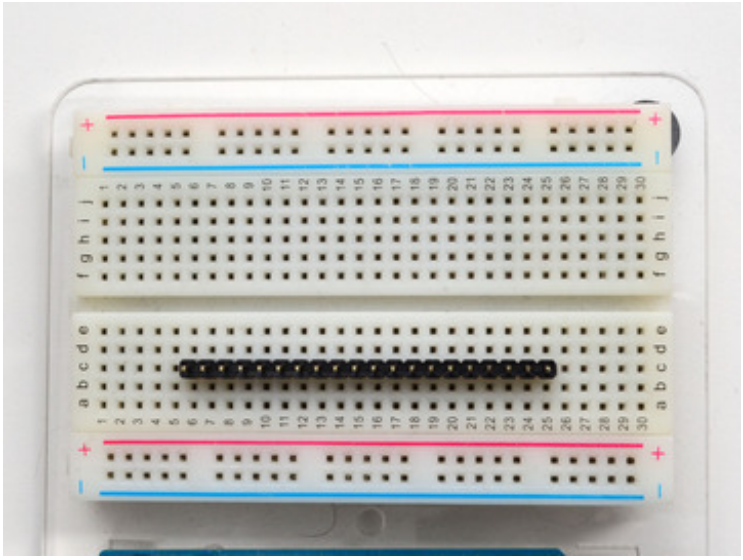


- **GND** - this is the power and signal ground pin
- **3-5V (Vin)**- this is the power pin, connect to 3-5VDC - it has reverse polarity protection but try to wire it right!
- **CS** - this is the TFT 8-bit chip select pin (it is also tied to the SPI mode **CS** pin)
- **C/D** - this is the TFT 8-bit data or command selector pin. It is **not the same as the SPI D/C pin!** Instead, it's the same as the SPI CLK pin.
- **WR** - this is the TFT 8-bit write strobe pin. It is also connected to the SPI **D/C** pin
- **RD** - this is the TFT 8-bit read strobe pin. You may not need this pin if you don't want to read data from the display
- **RST** - this is the TFT reset pin. There's auto-reset circuitry on the breakout so this pin is not required but it can be helpful sometimes to reset the TFT if your setup is not always resetting cleanly. Connect to ground to reset the TFT
- **Backkite** - this is the PWM input for the backlight control. It is by default pulled high (backlight on) you can PWM at any frequency or pull down to turn the backlight off
- **D0** thru **D7** - these are the 8 bits of parallel data sent to the TFT in 8-bit mode **D0** is the least-significant-bit and **D7** is the MSB

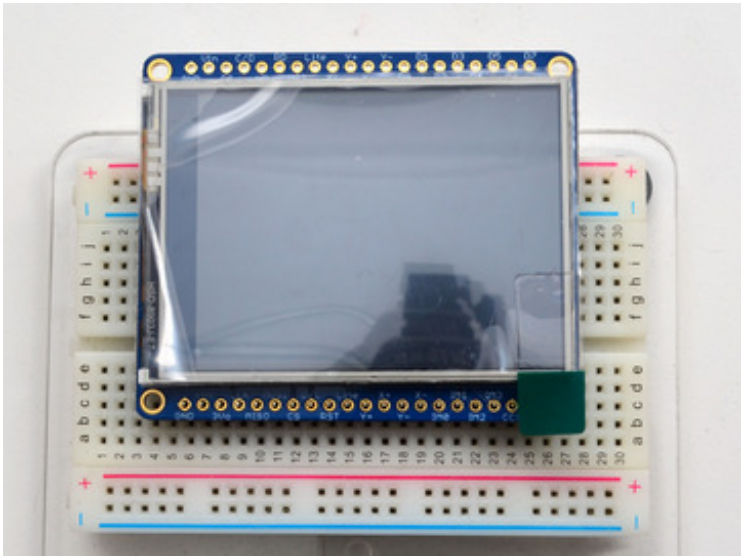
Assembly



Prepare the header strip:

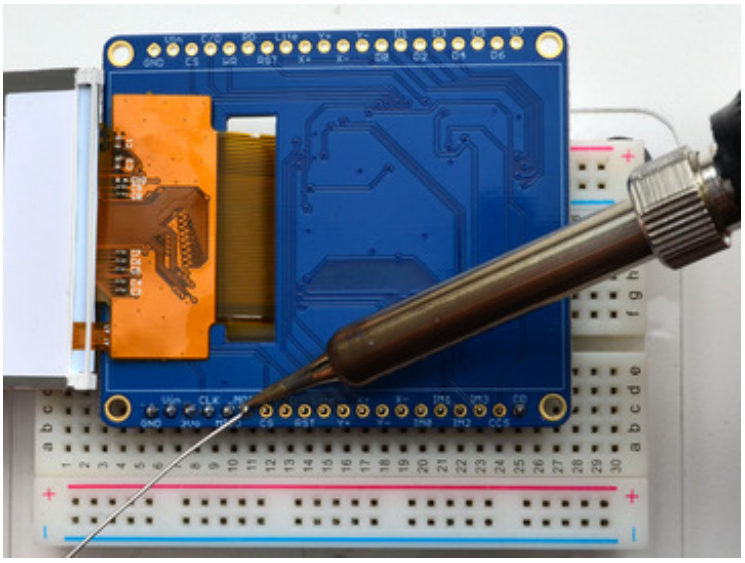
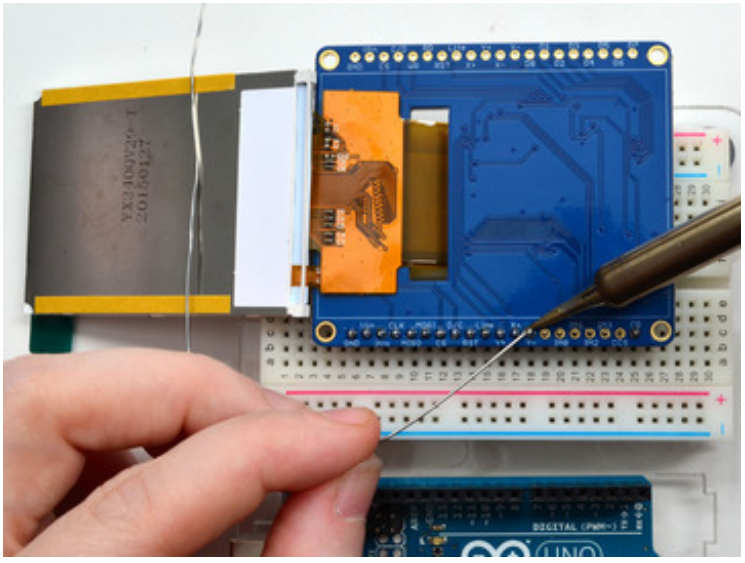
- 

Cut the strip to length if necessary. It will be easier to solder if you insert it into a breadboard - **long pins down**

- 

Add the breakout board:

Place the breakout board over the pins so that the short pins poke through the breakout pads

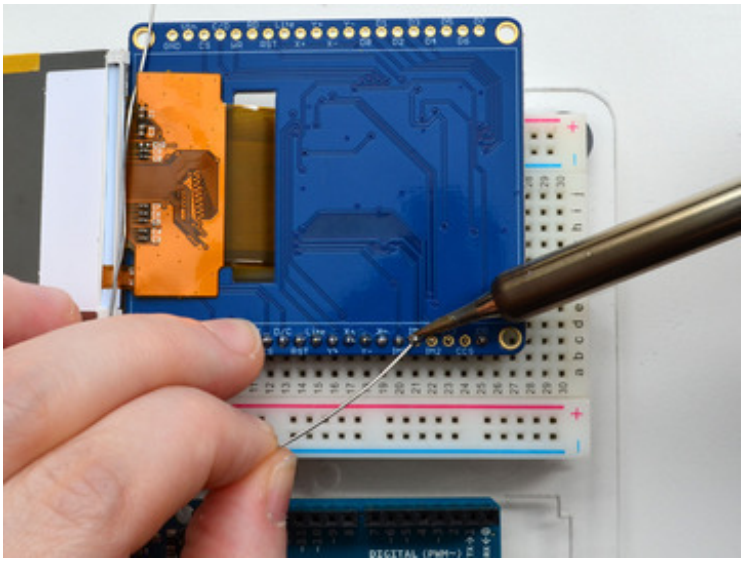
- 
- 

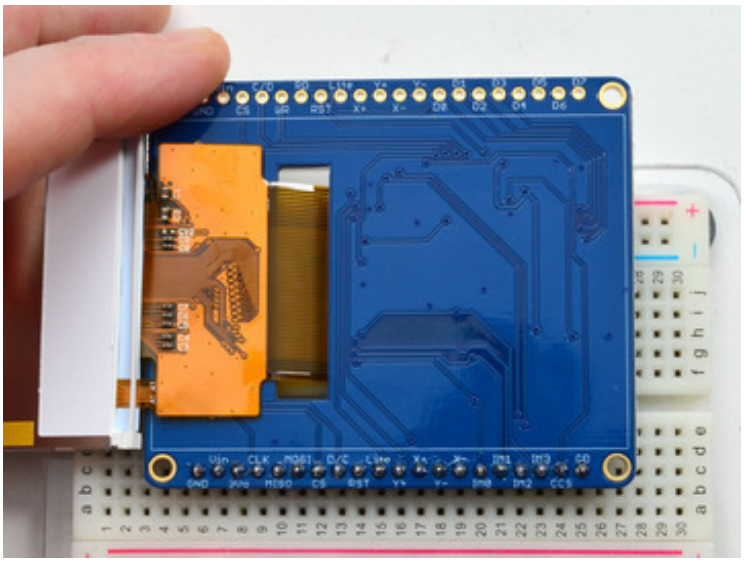
And Solder!

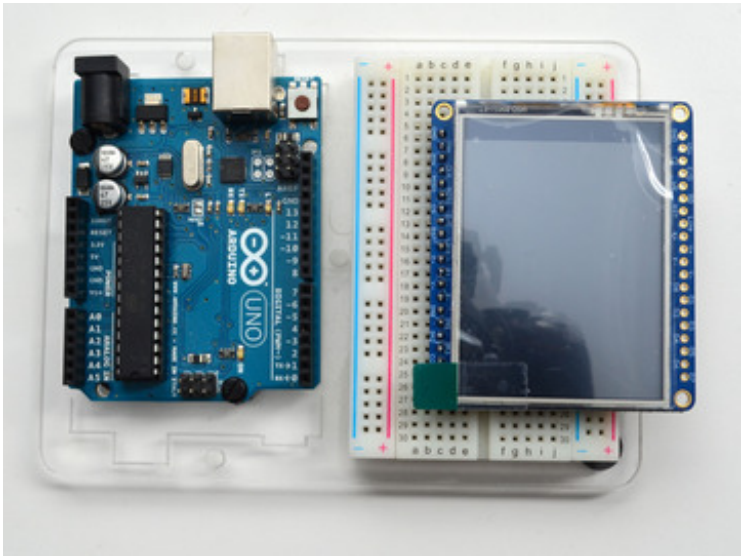
Be sure to solder all pins for reliable electrical contact.

Solder the longer power/data strip first

(For tips on soldering, be sure to check out our [Guide to Excellent Soldering](http://adafru.it/aTk) (http://adafru.it/aTk)).

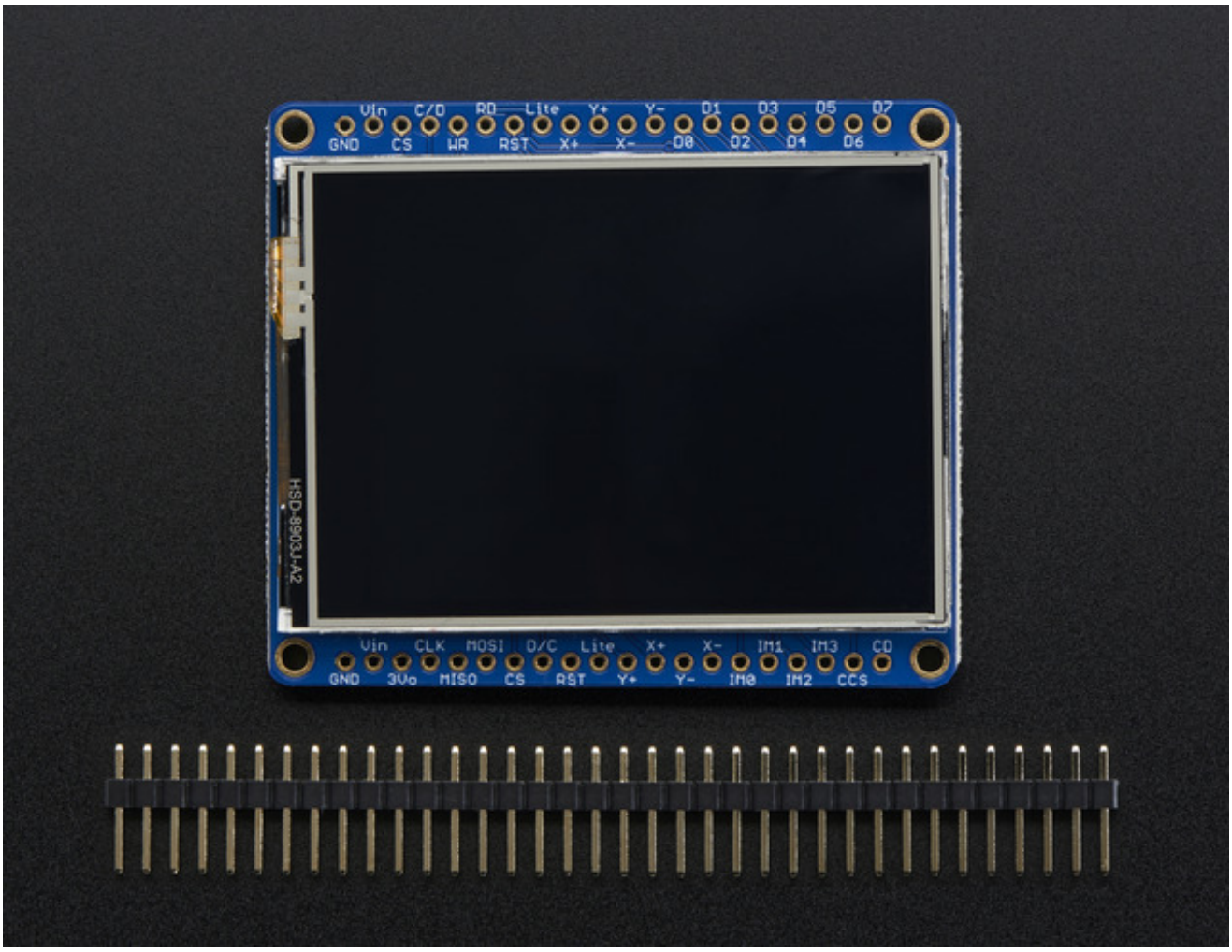
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You're done! Check your solder joints visually
- 

If your display isn't taped down to the breakout PCB, peel off the tape covering and press down so the TFT sticks to the circuit board

Wiring & Test



We tried to make this TFT breakout useful for both high-pin microcontrollers that can handle 8-bit data transfer modes as well as low-pincount micros like the Arduino UNO and Leonardo that are OK with SPI.

Essentially, the tradeoff is pins for speed. SPI is about 2-4 times slower than 8-bit mode, but that may not matter for basic graphics!

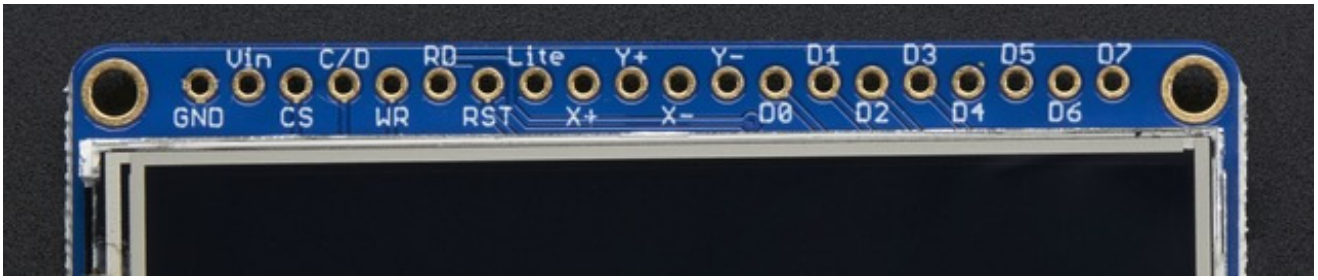
In addition, SPI mode has the benefit of being able to use the onboard microSD card socket for reading images. We don't have support for this in 8-bit mode so if you want to have an all-in-one image viewer type application, use SPI!

8-Bit Wiring & Test

8-Bit Wiring

Wiring up the 8-bit mode is kind of a pain, so we really only recommend doing it for UNO (which we show) and Mega (which we describe, and is pretty easy since its 8 pins in a row). Anything else, like a Leonardo or Micro, we strongly recommend going with SPI mode since we don't have an example for that. Any other kind of 'Arduino compatible' that isn't an Uno, try SPI first. The 8-bit mode is hand-tweaked in the **Adafruit_TFTLCD pin_magic.h** file. Its really only for advanced users who are totally cool with figuring out bitmasks for various ports & pins.

Really, we'll show how to do the UNO but anything else? go with SPI!



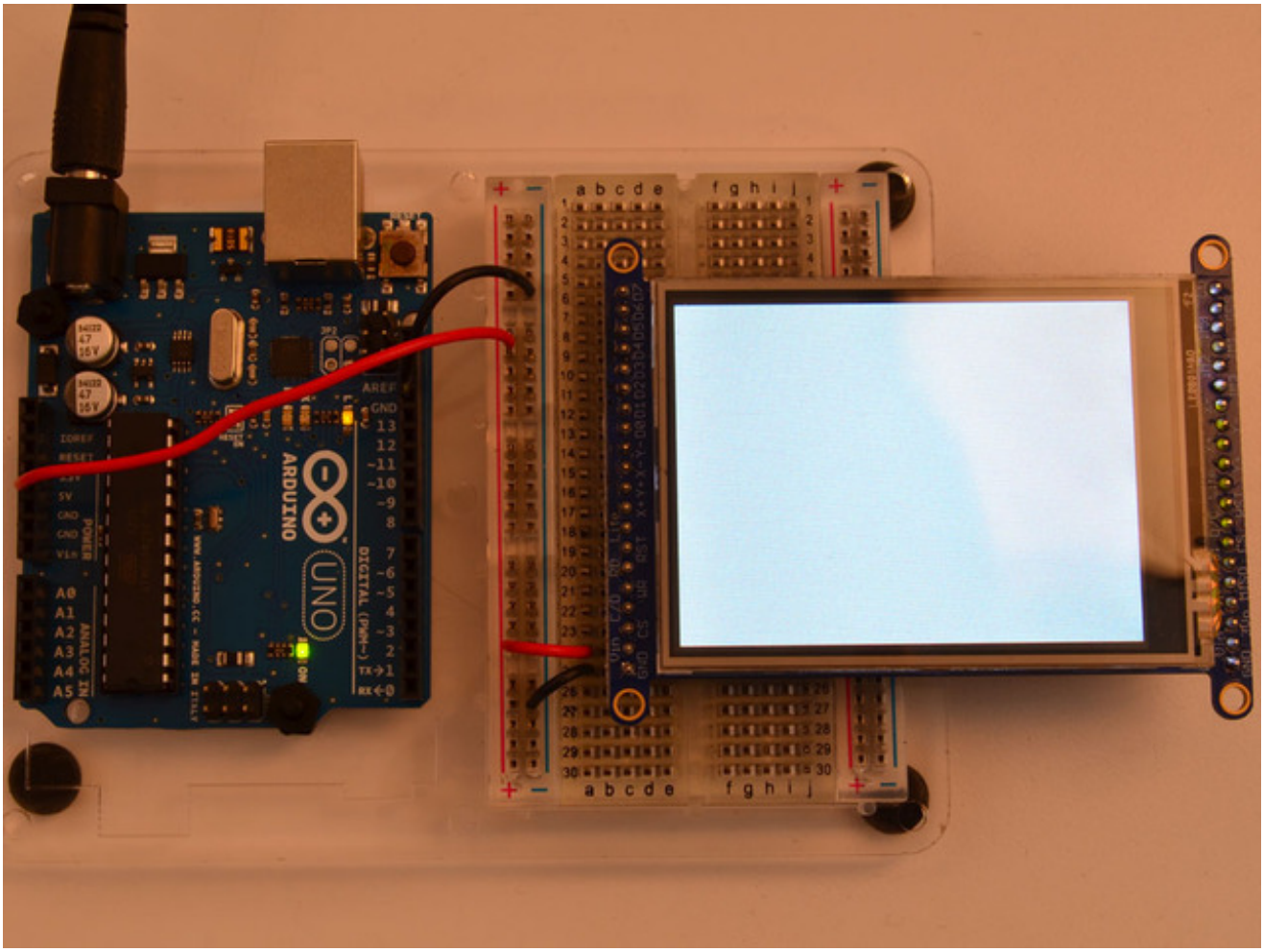
Make sure you're soldering and connecting to the 8-bit side!

Part 1 - Power & backlight test

Begin by wiring up the **3-5VDC** and **GND** pins.

Connect the **3-5V** pin to **5V** and **GND** to **GND** on your Arduino. I'm using the breadboard rails but you can also just wire directly.

Power it up and you should see the white backlight come on.



Part 2 - Data Bus Lines

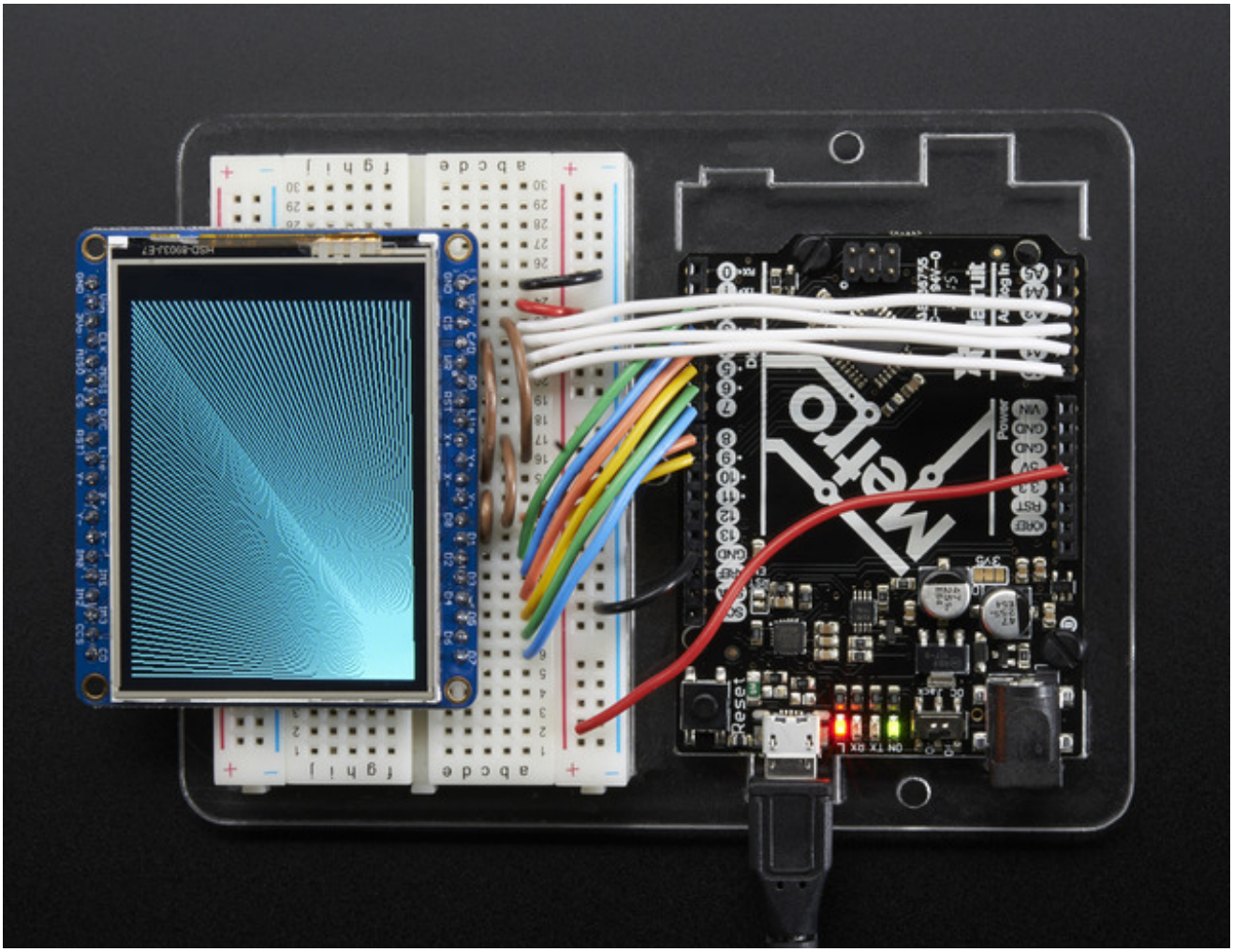
Now that the backlight is working, we can get the TFT LCD working. There are many pins required, and to keep the code running fairly fast, we have 'hardcoded' Arduino digital pins **#2-#9** for the 8 data lines.
However, they are not in that order! D0 and D1 go to **digital #8** and **#9**, then D2-D7 connect to **#2** thru **#7**. This is because Arduino pins #0 and #1 are used for serial data so we can't use them

Begin by connecting D0 and D1 to digital #8 and 9 respectively as seen above. If you're using a Mega, connect the TFT Data Pins D0-D1 to Mega pins #22-23, in that order. Those Mega pins are on the 'double' header.

Now you can connect the remaining 6 pins over. Connect **D2-D7** on the TFT pins to digital **2** thru **7** in that order. If you're using a Mega, connect the TFT Data Pins **D2-D7** to Mega pins **#24-29**, in that order. Those Mega pins are on the 'double' header.

In addition to the 8 data lines, you'll also need 4 or 5 control lines. These can later be reassigned to any digital pins, they're just what we have in the tutorial by default.

- Connect the third pin **CS** (**Chip Select**) to Analog 3
- Connect the fourth pin **C/D** (**Command/Data**) to Analog 2
- Connect the fifth pin **WR** (**Write**) to Analog 1
- Connect the sixth pin **RD** (**Read**) to Analog 0



You can connect the seventh pin **RST** (**Reset**) to the Arduino Reset line if you'd like. This will reset the panel when the Arduino is Reset. You can also use a digital pin for the LCD reset if you want to manually reset. There's auto-reset circuitry on the board so you probably don't need to use this pin at all and leave it disconnected

The **RD** pin is used to read the chip ID off the TFT. Later, once you get it all working, you can remove this pin and the ID test, although we suggest keeping it since its useful for debugging your wiring. OK! Now we can run some code

8-Bit Library Install

We have example code ready to go for use with these TFTs. It's written for Arduino, which should be portable to any microcontroller by adapting the C++ source.

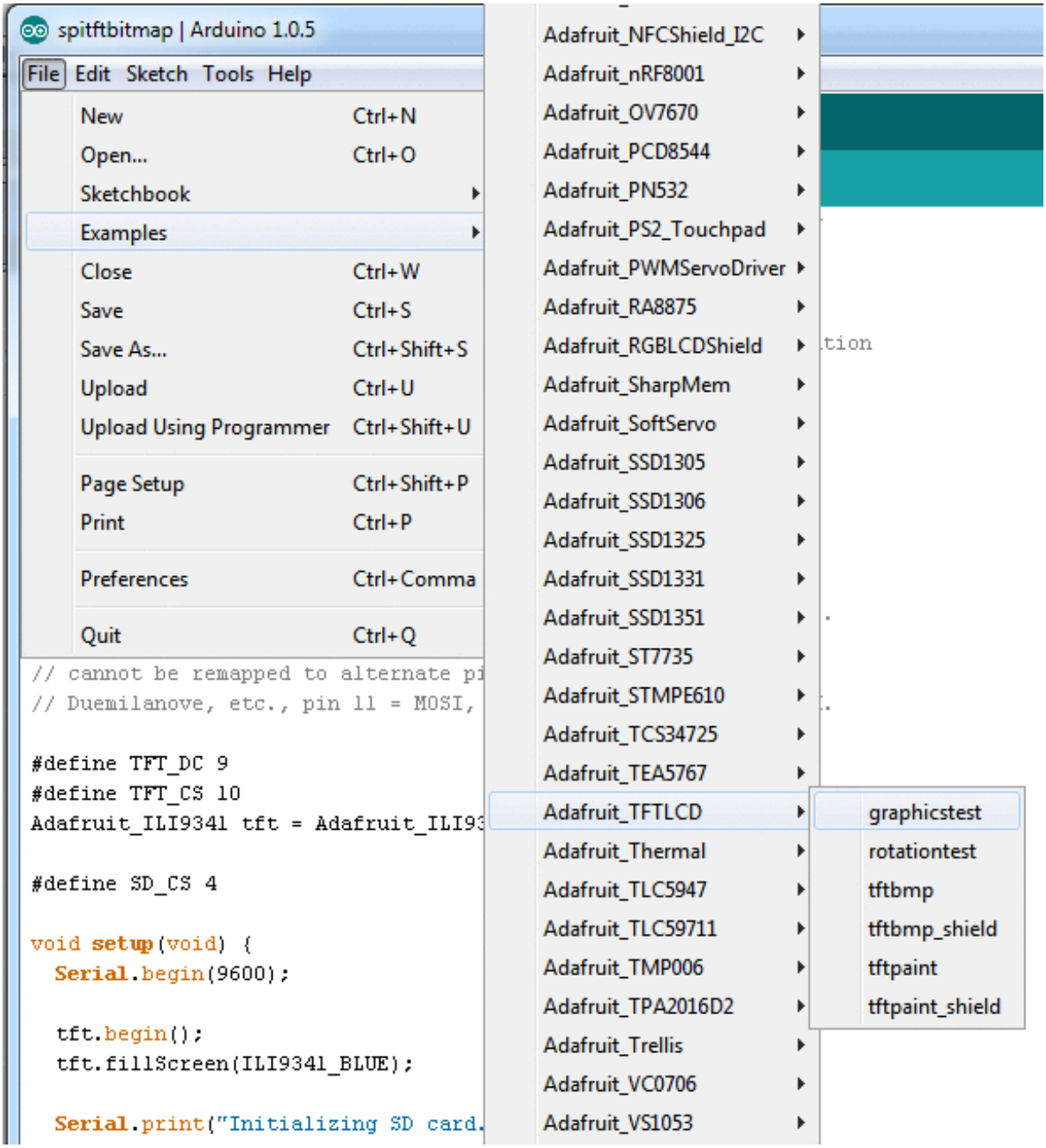
Two libraries need to be downloaded and installed: first is theAdafruit_TFTLCD library (http://adafru.it/aHk) (this contains the low-level code specific to this device), and second is the Adafruit GFX Library (http://adafru.it/aJa) (which handles graphics operations common to many displays we carry). If you have **Adafruit_GFX** already, make sure its the most recent version since we've made updates for better performance

Download the Adafruit TFTLCD Library
http://adafru.it/dcW
Download the Adafruit GFX Library
http://adafru.it/cBB

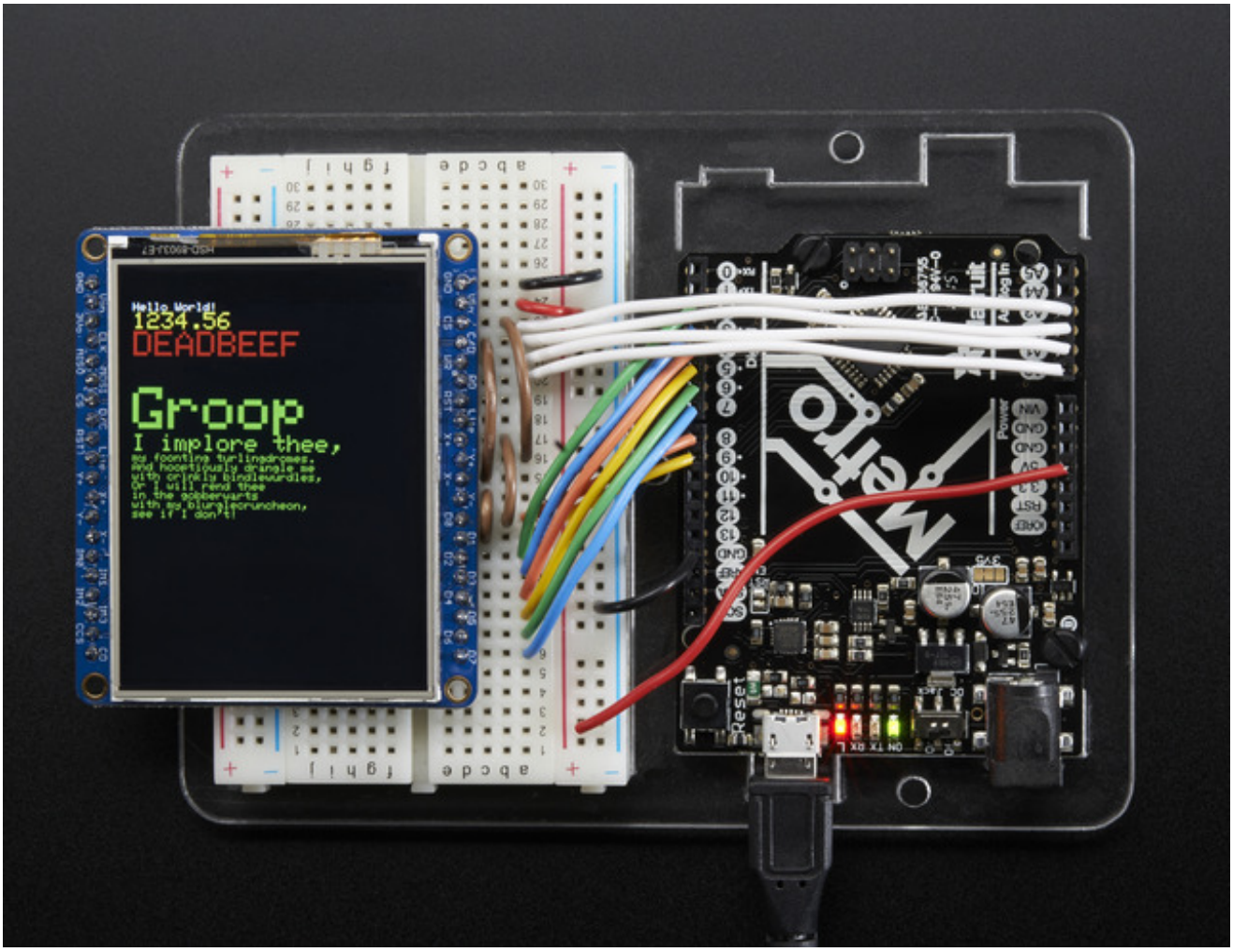
Download both ZIP files, uncompress and rename the folders to**Adafruit_TFTLCD** (contains **Adafruit_TFTLCD.cpp** and **.h**) and **Adafruit_GFX** (contains **Adafruit_GFX.cpp** and **.h**) respectively. Then place them inside your Arduino**libraries** folder and restart the Arduino IDE. If this is all unfamiliar, we have a tutorial introducing Arduino library concepts and installation (http://adafru.it/aYM).
In the **Adafruit_TFTLCD** Library folder, you may need to edit**Adafruit_TFTLCD.h**. On about line 12, you will see

```
#define USE_ADAFRUIT_SHIELD_PINOUT
```

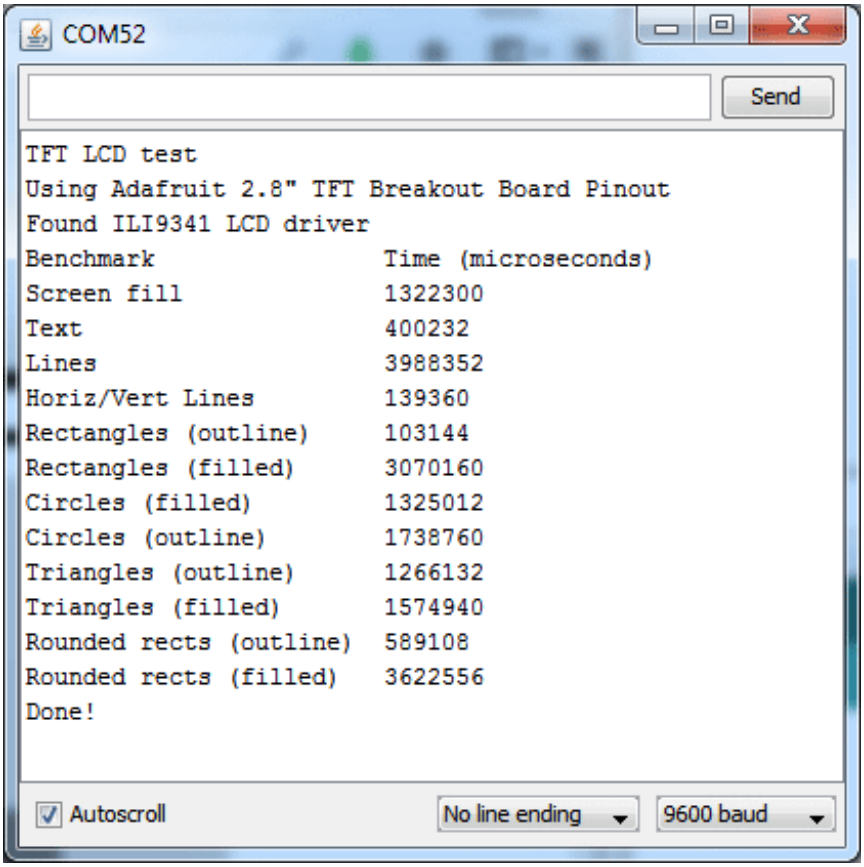
Make sure this line is commented out with a// in front (it should but if you're having issues, its worth checking.)



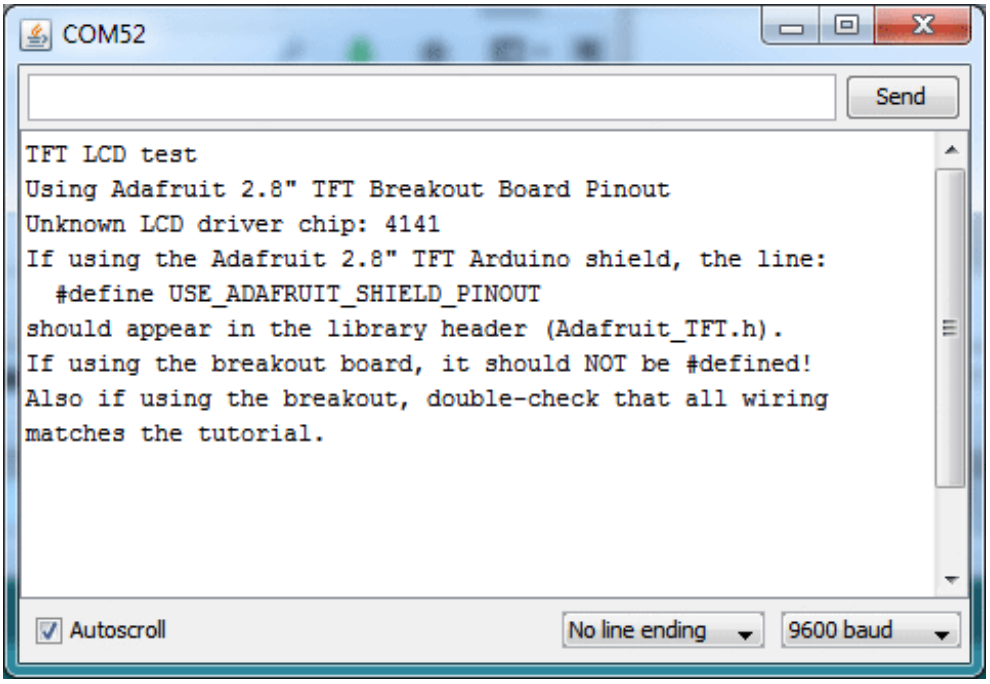
After restarting the Arduino software, you should see a new **example** folder called **Adafruit_TFTLCD** and inside, an example called **graphicstest**. Upload that sketch to your Arduino. You may need to press the Reset button to reset the arduino and TFT. You should see a collection of graphical tests draw out on the TFT.



If you're having difficulties, check the serial console.The first thing the sketch does is read the driver code from the TFT. It should be **0x9341** (for the **ILI9341** controller inside)



If you **Unknown Driver Chip** then it's probably something with your wiring, double check and try again!



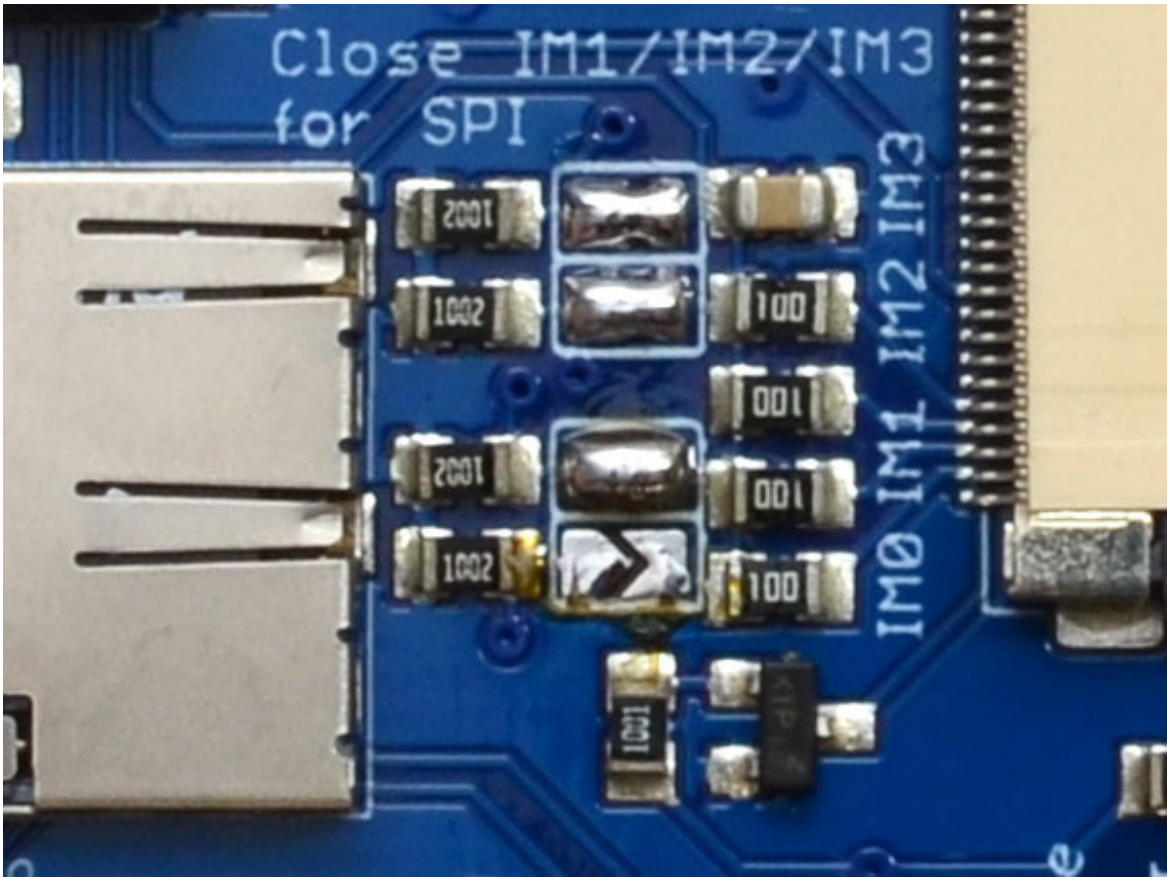
SPI Wiring & Test



Don't forget, we're using the SPI interface side of the PCB!

SPI Mode Jumpers

Before you start, we'll need to tell the display to put us in SPI mode so it will know which pins to listen to. To do that, we have to connect tbe **IM1**, **IM2** and **IM3** pins to 3.3V. The easiest way to do that is to solder closed the **IMx** jumpers on the back of the PCB. Turn over the PCB and find the solder jumpers



With your soldering iron, melt solder to close the three jumpers indicated**IM1 IM2** and **IM3**

(do not solder closed **IM0**!)

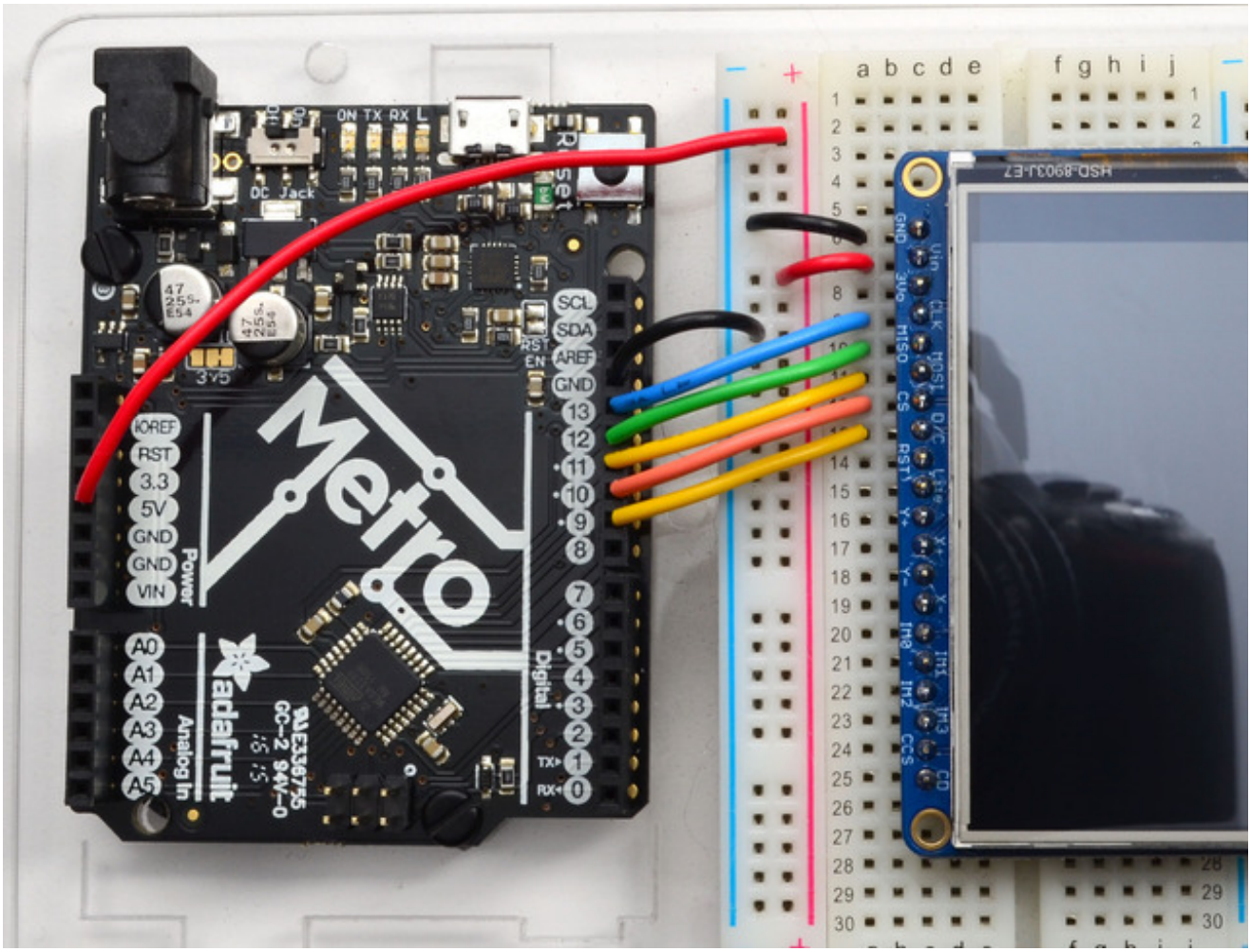
If you really don't want to solder them, you can also wire the breakout pins to the **3vo** pin, just make sure you don't tie them to 5V by accident! For that reason, we suggest going with the solder-jumper route.

Wiring

Wiring up the display in SPI mode is much easier than 8-bit mode since there's way fewer wires. Start by connecting the power pins

- **3-5V Vin** connects to the Arduino **5V** pin
- **GND** connects to Arduino ground
- **CLK** connects to SPI clock. On Arduino Uno/Duemilanove/328-based, thats **Digital 13**. On Mega's, its **Digital 52** and on Leonardo/Due its **ICSP-3** (See SPI Connections for more details (http://adafru.it/d5h))
- **MISO** connects to SPI MISO. On Arduino Uno/Duemilanove/328-based, thats **Digital 12**. On Mega's, its **Digital 50** and on Leonardo/Due its **ICSP-1** (See SPI Connections for more details (http://adafru.it/d5h))
- **MOSI** connects to SPI MOSI. On Arduino Uno/Duemilanove/328-based, thats **Digital 11**. On Mega's, its **Digital 51** and on Leonardo/Due its **ICSP-4** (See SPI Connections for more details (http://adafru.it/d5h))
- **CS** connects to our SPI Chip Select pin. We'll be using **Digital 10** but you can later change this to any pin
- **D/C** connects to our SPI data/command select pin. We'll be using **Digital 9** but you can later change this pin too.

That's it! You do not need to connect the **RST** or other pins for now.



Install Adafruit ILI9341 TFT Library

We have example code ready to go for use with these TFTs. It's written for Arduino, which should be portable to any microcontroller by adapting the C++ source.

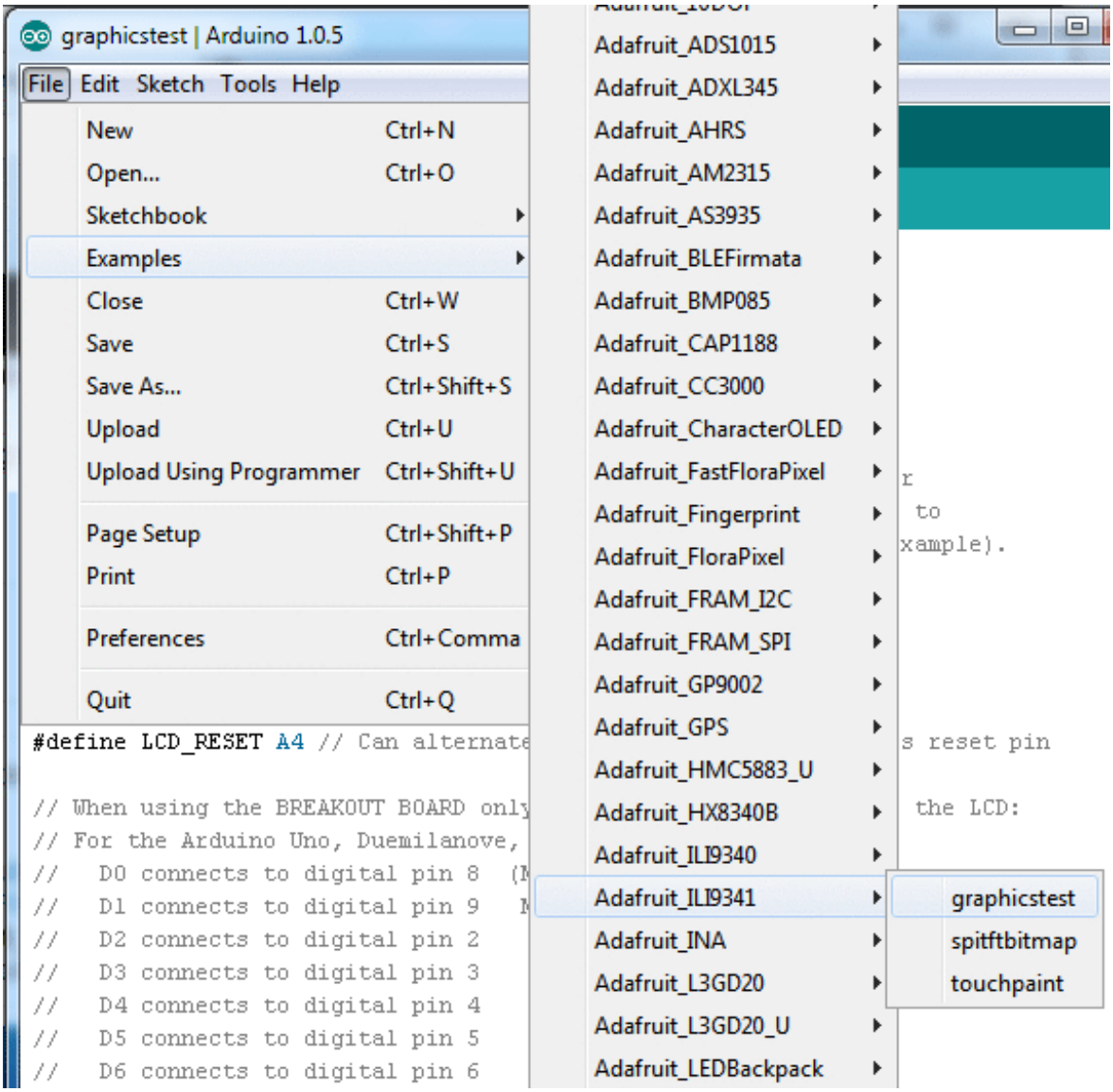
Two libraries need to be downloaded and installed: first is the[Adafruit_ILI9341 library](http://adafru.it/d4d) (http://adafru.it/d4d) (this contains the low-level code specific to this device), and second is the [Adafruit GFX Library](http://adafru.it/aJa) (http://adafru.it/aJa) (which handles graphics operations common to many displays we carry). If you have **Adafruit_GFX** already, make sure its the most recent version since we've made updates for better performance

[Download the latest Adafruit_ILI9341 Library](http://adafru.it/d4e)
http://adafru.it/d4e
[Download the latest Adafruit_GFX Library](http://adafru.it/cBB)
http://adafru.it/cBB

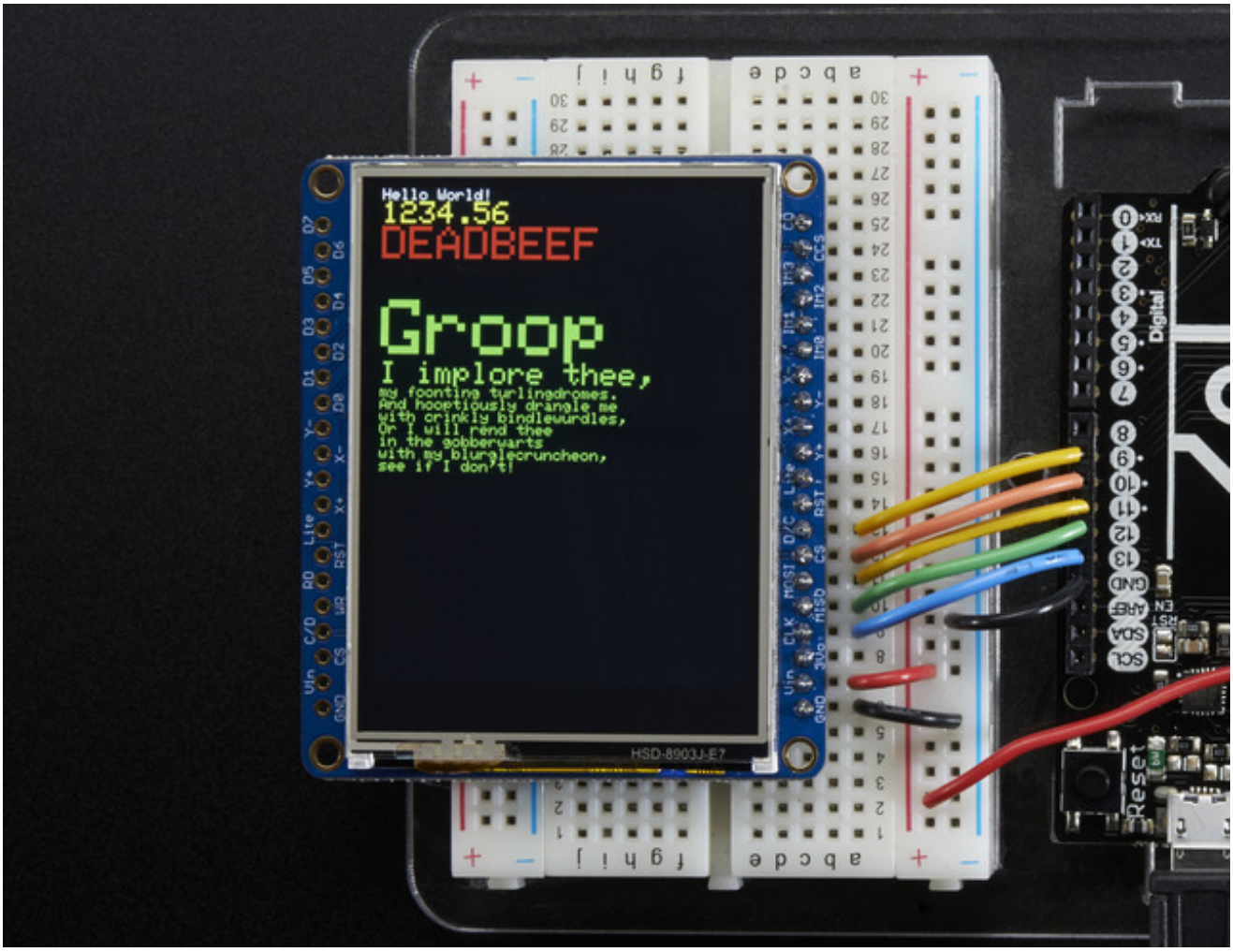
Download both ZIP files, uncompress and rename the folders to**Adafruit_ILI9341**

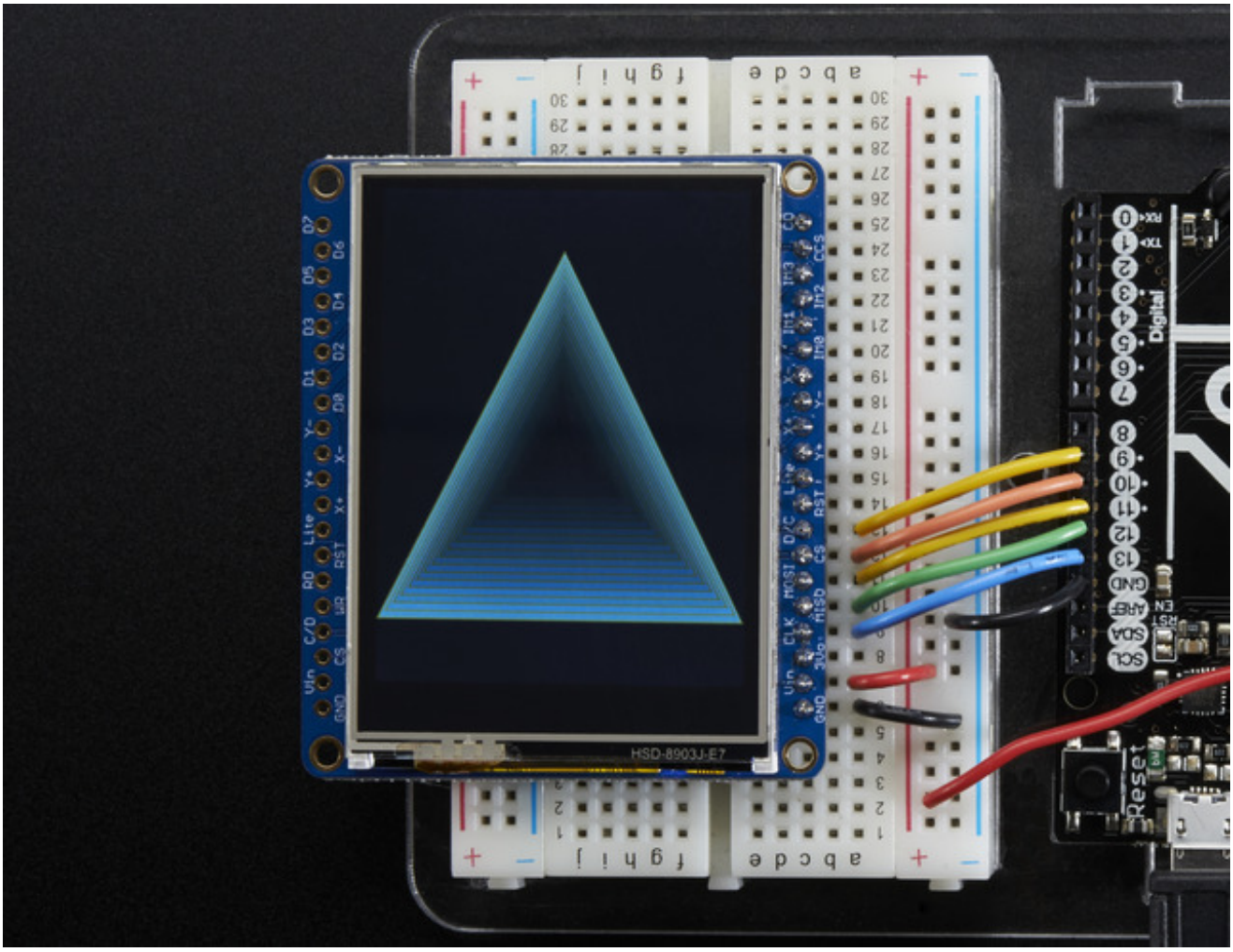
(contains **Adafruit_ILI9341.cpp** and **.h**) and **Adafruit_GFX** (contains **Adafruit_GFX.cpp** and **.h**) respectively. Then place them inside your Arduino **libraries** folder and restart the Arduino IDE. If this is all unfamiliar, we have a [tutorial introducing Arduino library concepts and installation](http://adafru.it/aYM) (http://adafru.it/aYM).

Restart the IDE



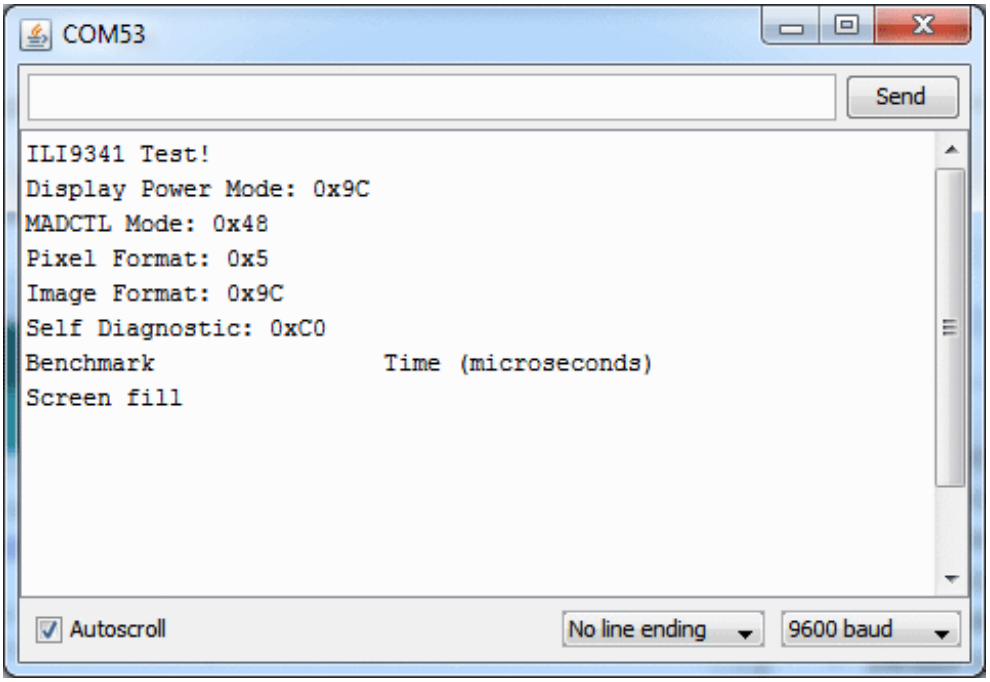
After restarting the Arduino software, you should see a new **example** folder called **Adafruit_ILI9341** and inside, an example called **graphicstest**. Upload that sketch to your Arduino. You may need to press the Reset button to reset the arduino and TFT. You should see a collection of graphical tests draw out on the TFT.


Hello World!
1234.56
DEADBEEF
Groop
I implore thee,
my foonting turlingdromes.
And hooptiously drangle me
with crinkly bindlewurdles,
Or I will rend thee
in the gobbervarts
with my blurglecruncheon,
see if I don't!
HSD-8903J-E7
Digital
Reset



If you're having difficulties, check the serial console.The first thing the sketch does is read the driver configuration from the TFT, you should see the same numbers as below

If you did not connect up the MISO line to the TFT, you wont see the read configuation bytes so please make sure you connect up the MISO line for easy debugging! Once its all working, you can remove the MISO line


COM53
Send
ILI9341 Test!
Display Power Mode: 0x9C
MADCTL Mode: 0x48
Pixel Format: 0x5
Image Format: 0x9C
Self Diagnostic: 0xC0
Benchmark Time (microseconds)
Screen fill
Autoscroll
No line ending
9600 baud

Bitmaps (SPI Mode)

There is a built in microSD card slot into the breakout, and we can use that to load bitmap images! You will need a microSD card formatted **FAT16 or FAT32** (they almost always are by default).

Its really easy to draw bitmaps. **However, this is only supported when talking to the display in SPI mode, not 8-bit mode!**

Lets start by downloading this image of pretty flowers

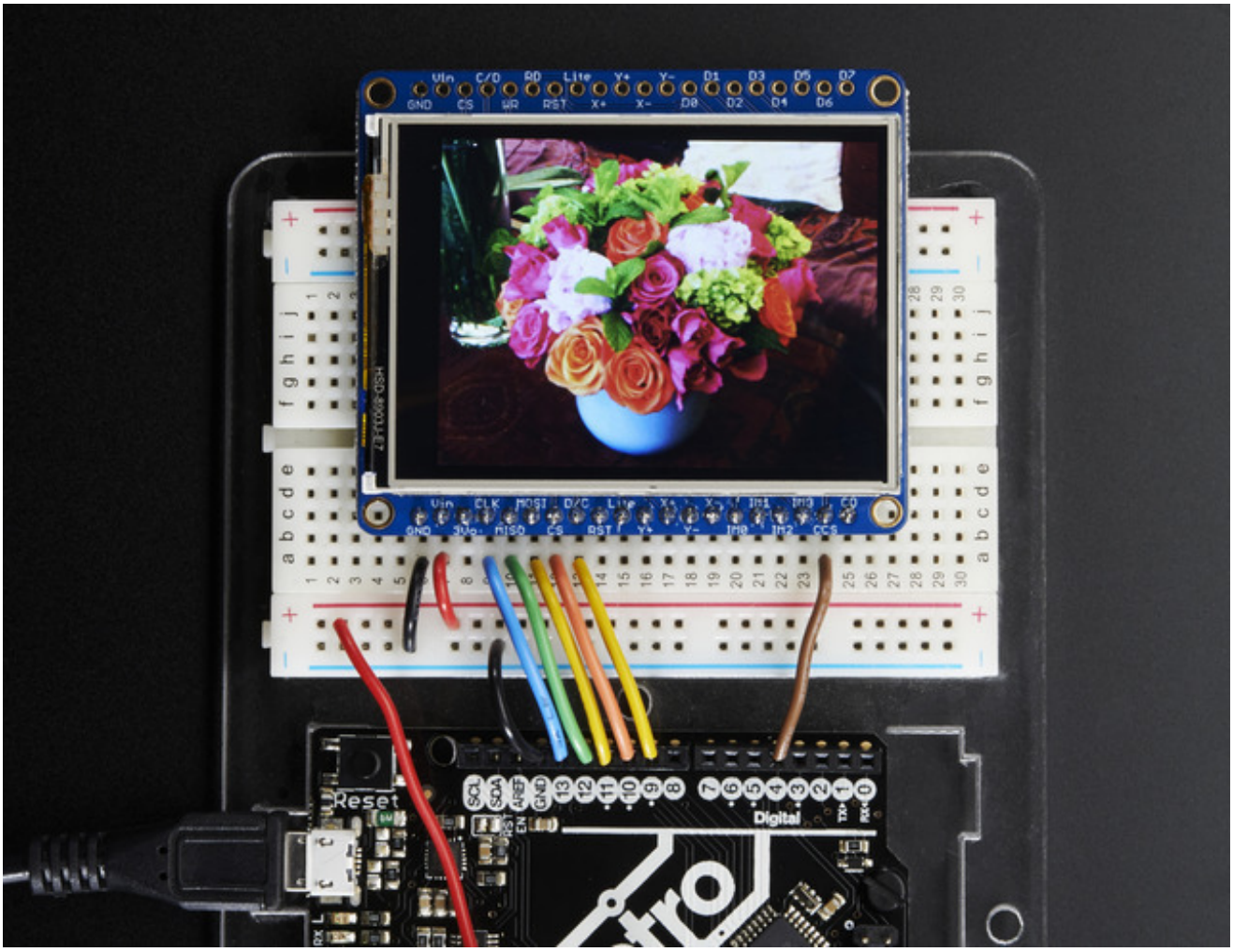


Copy **purple.bmp** into the base directory of a microSD card and insert it into the microSD socket in the breakout.

You'll need to connect up the **CCS** pin to **Digital 4** on your Arduino as well. In the below image, its the extra brown wire

You may want to try the **SD library** examples before continuing, especially one that lists all the files on the SD card

Now upload the **file->examples->Adafruit_ILI9341->spitftbitmap** example to your Arduino + breakout. You will see the flowers appear!



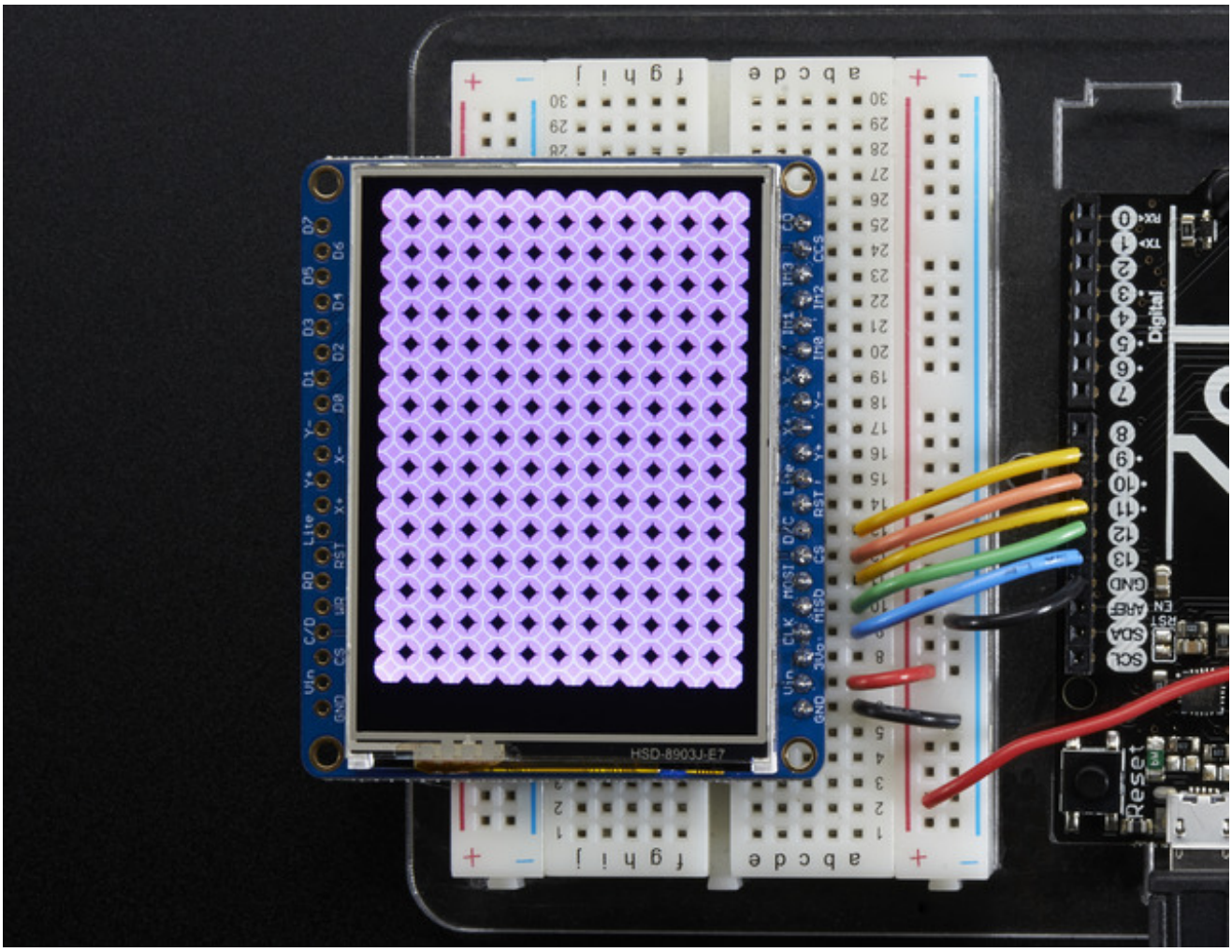
Copy **purple.bmp** into the base directory of a microSD card and insert it into the microSD socket in the breakout.

You'll need to connect up the **CCS** pin to **Digital 4** on your Arduino as well. In the below image, its the extra purple wire

You may want to try the **SD library** examples before continuing, especially one that lists all the files on the SD card

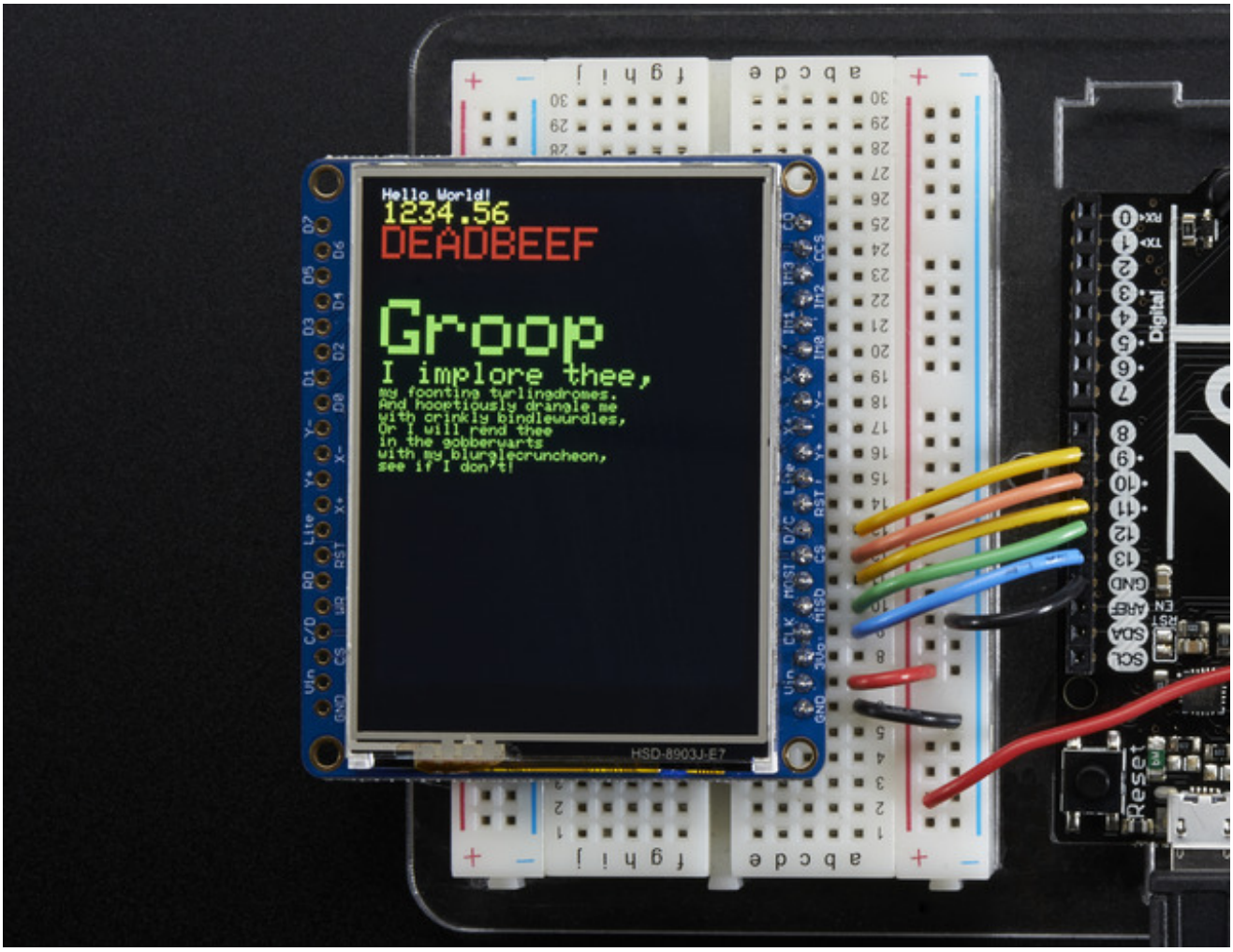
Now upload the **file->examples->Adafruit_ILI9341->spitftbitmap** example to your Arduino + breakout. You will see the flowers appear!

Adafruit GFX Library



The Adafruit_GFX library for Arduino provides a common syntax and set of graphics functions for all of our TFT, LCD and OLED displays. This allows Arduino sketches to easily be adapted between display types with minimal fuss…and any new features, performance improvements and bug fixes will immediately apply across our complete offering of color displays.

The GFX library is what lets you draw points, lines, rectangles, round-rects, triangles, text, etc.



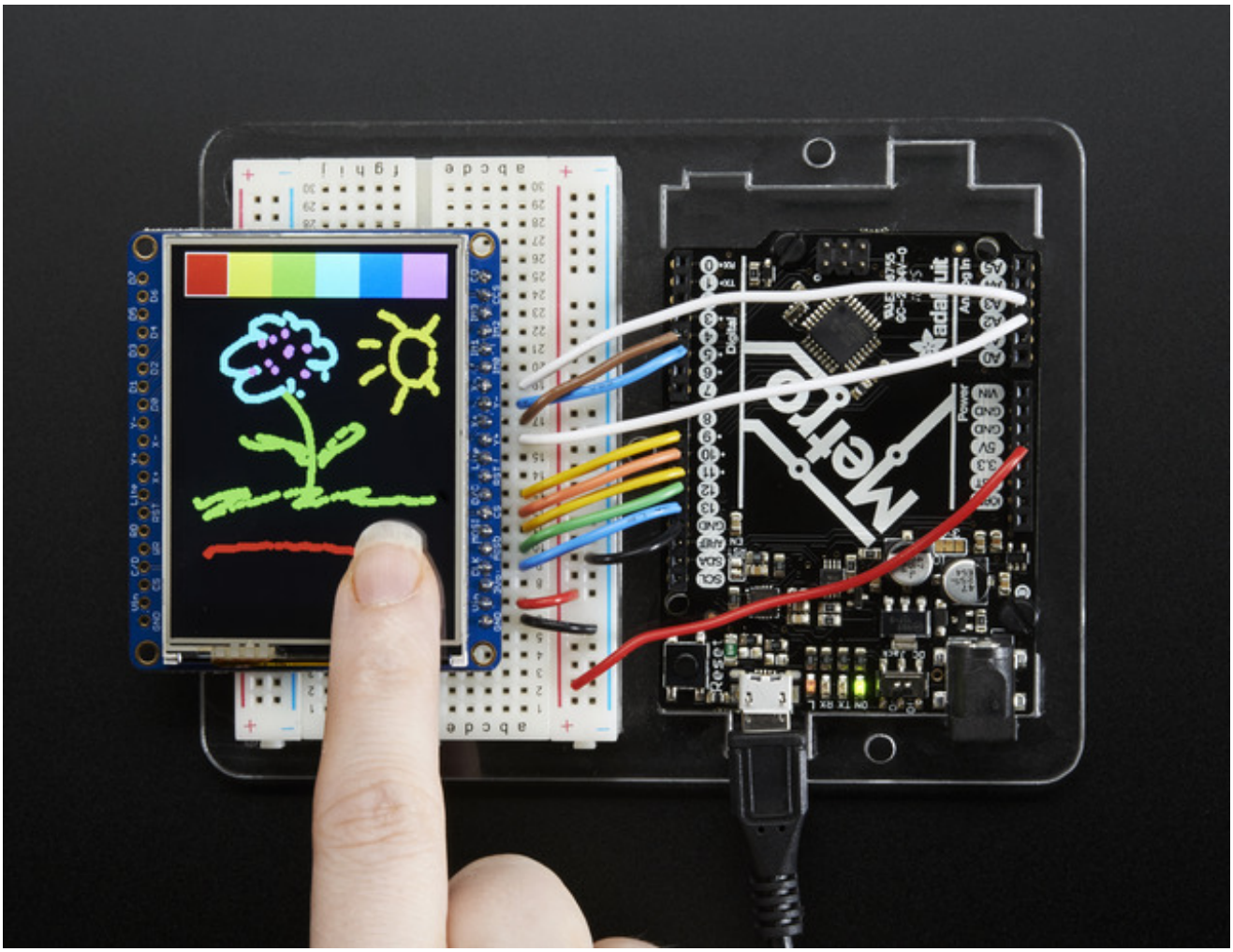
Check out our detailed tutorial here http://learn.adafruit.com/adafruit-gfx-graphics-library (http://adafru.it/aPx)

It covers the latest and greatest of the GFX library. The GFX library is used in both 8-bit and SPI modes so the underlying commands (drawLine() for example) are identical!

Resistive Touchscreen

The LCD has a 2.8" 4-wire resistive touch screen glued onto it. You can use this for detecting finger-presses, stylus', etc. You'll need 4 pins to talk to the touch panel, and at least 2 must be analog inputs. The touch screen is a completely separate part from the TFT, so be aware if you rotate the display or have the TFT off or reset, the touch screen doesn't "know" about it - its just a couple resistors!

We have a demo for the touchscreen + TFT that lets you 'paint' simple graphics. There's versions for both SPI and 8-bit mode and are included in the libraries. Just make sure you have gone thru the TFT test procedure already since this builds on that.



Remember, if you rotate the screen drawing with setRotation() you'll have to use map() or similar to flip around the X/Y coordinates for the touchscreen as well! It doesn't know about drawing rotation

Download Library

Begin by grabbing our analog/resistive touchscreen library from github (http://adafru.it/aT1) (or just click the download button)

Download resistive touchscreen library
http://adafru.it/dd0

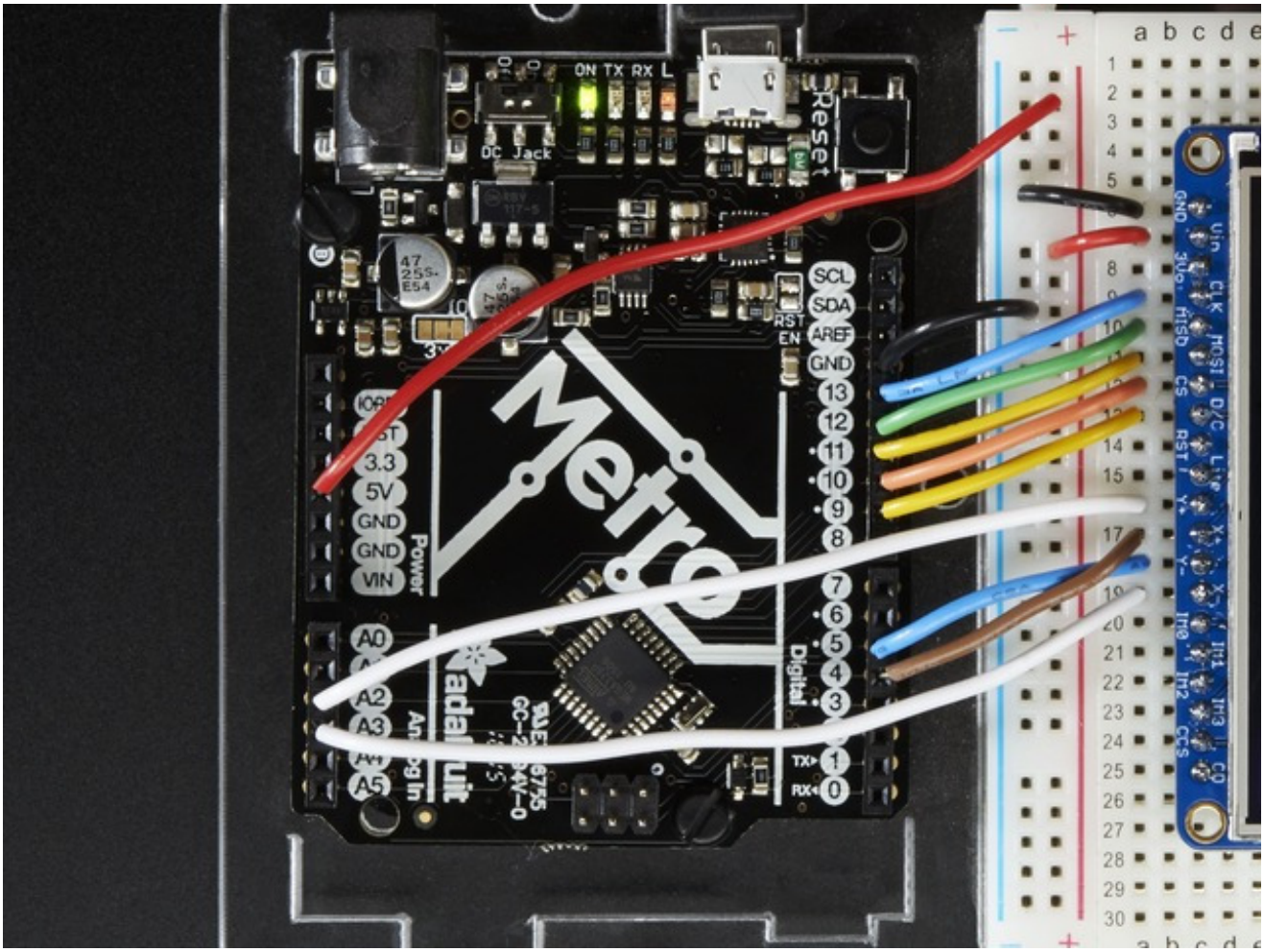
Uncompress the zip file and rename the folder **TouchScreen** (make sure it contains **Touchscreen.cpp** and **Touchscreen.h**) then install in your **libraries** folder just like you did for the TFT library

Touchscreen Paint (SPI mode)

An additional 4 pins are required for the touchscreen. For the two analog pins, we'll use **A2** and **A3**. For the other two connections, you can pin any two digital pins. We can save the one pin by sharing with **D**/**C** but you can't share any other SPI pins. So basically you can get away with using only three additional pins.

Wire the additional 4 pins as follows:

- Y+ to Arduino **A2**
- X+ to Arduino **D4**
- Y- to Arduino **D5**
- X- to Arduino **A3**



Load up the **breakoutTouchPaint** example from the **Adafruit_ILI9341** library and try drawing with your fingernail! You can select colors by touching the 'pallette' of colors on top

Touchscreen Paint (8-Bit mode)

Another 4 pins seems like a lot since already 12 are taken up with the TFT **but** you can **reuse** some of the pins for the TFT LCD! This is because the resistance of the panel is high enough that it doesn't interfere with the digital input/output and we can query the panel in between TFT accesses, when the pins are not being used.

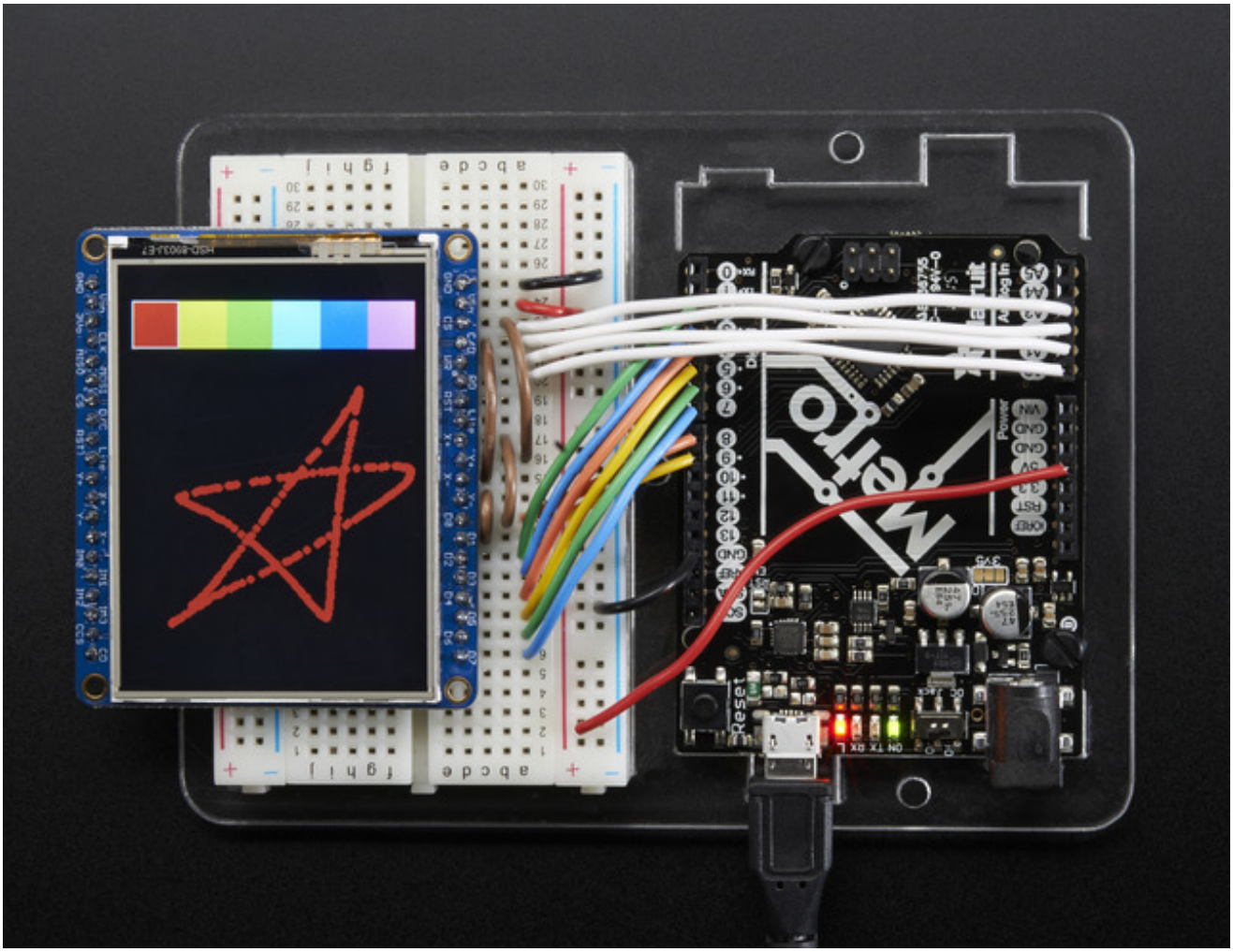
We'll be building on the wiring used in the previous drawing test for UNO

You can wire up the 4 touchscreen pins as follows. Starting from the top

- **Y-** connects to digital **#9** (also **D1**)
- The next one down (**X-**) connects to **Analog 2** (also **C/D**)

- The next one over (**Y+**) connects to **Analog 3** (also **CS**)
- The last one (**X+**) connects to digital **8**. (also **D0**)

The X- and Y+ pins pretty much have to connect to those analog pins (or to analog 4/5) but Y-/X+ can connect to any digital or analog pins.



Load up the **tftpaint** example from the **Adafruit_TFTLCD** library and try drawing with your fingernail! You can select colors by touching the 'pallette' of colors on the right

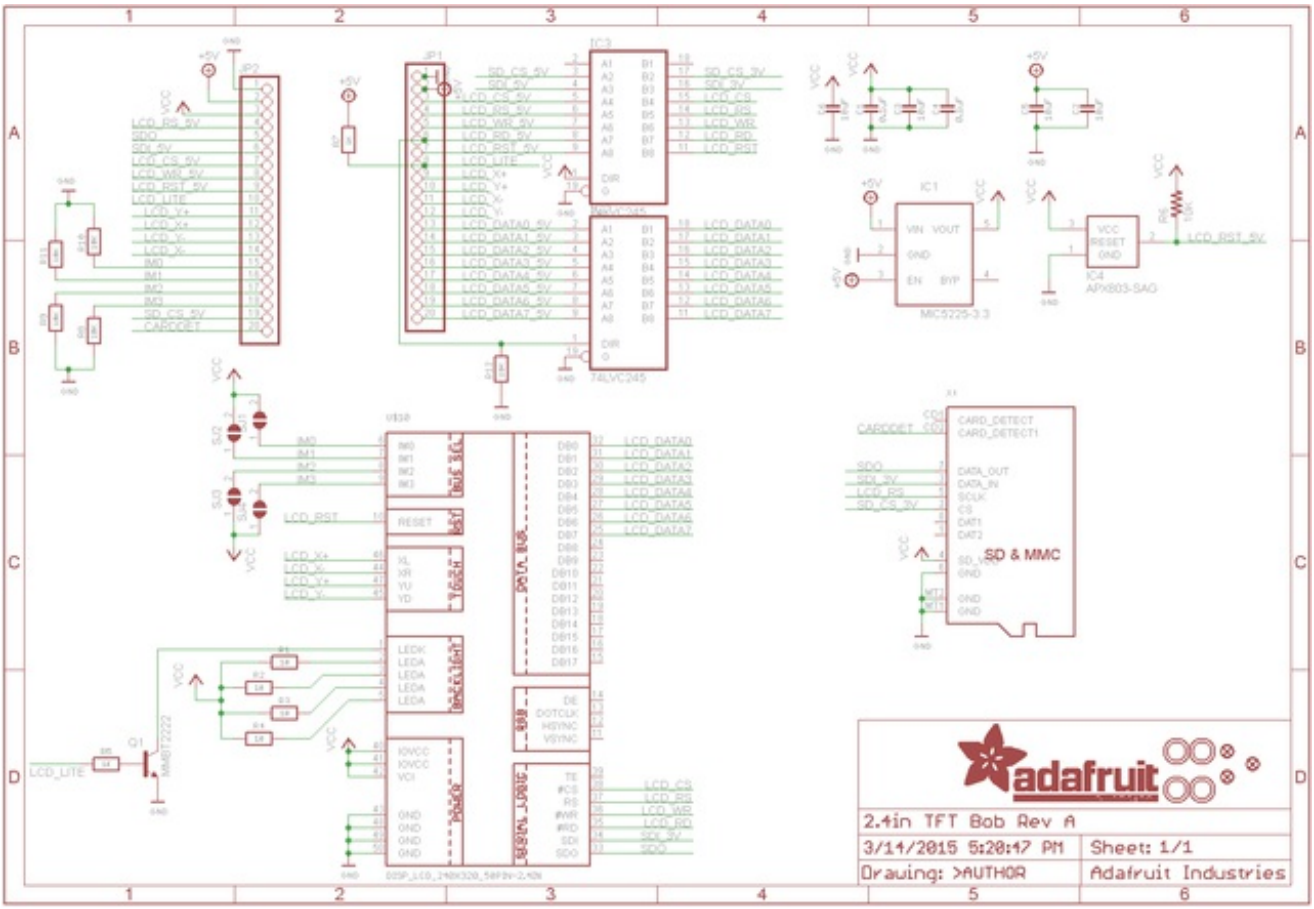
Downloads

Datasheets

- ILI9341 TFT controller chip datasheet (http://adafru.it/d4l) (this is what you want to refer to if porting or if you want to look at the TFT command set)
- Datasheet for the raw 2.4" screen itself (http://adafru.it/fgr)
- Adafruit ILI9341 Arduino driver library (http://adafru.it/d4d)
- EagleCAD PCB files on GitHub (http://adafru.it/rEl)
- Fritzing object in the Adafruit Fritzing Library (http://adafru.it/aP3)

Schematic

Click to embiggen



Fabrication Print

Dimensions in Inches

